

Quadratic Strong-Operator Paving over Arbitrary Maximal Abelian Subalgebras

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Abstract

We prove the quadratic strong-operator paving conjecture of Popa and Vaes. For every $0 < \varepsilon < 1$, every self-adjoint element of a von Neumann algebra admits strong-operator paving over each maximal abelian subalgebra with at most $5 \times 10^8 \varepsilon^{-2}$ projections. The bound is uniform over representations and requires no separability or conditional-expectation assumption.

1 Introduction

Let $M \subseteq \mathcal{B}(H)$ be a von Neumann algebra on a complex Hilbert space, and let $A \subseteq M$ be a maximal abelian unital $*$ -subalgebra, or *masa*. A finite partition $P = (p_1, \dots, p_r)$ in A consists of mutually orthogonal projections with sum 1. Its associated pinching is

$$S_P(x) = \sum_{i=1}^r p_i x p_i, \quad x \in M.$$

Paving asks how small this operator can be made after subtracting a diagonal element of A . Requiring a small norm for $S_P(x) - a$ itself is norm paving, which fails for general masas [PV15, Theorem 3.3]. The following strong-operator formulation instead permits a compression whose complement is arbitrarily small in the strong operator topology.

Definition 1.1 (Strong-operator paving). Let $x = x^* \in M$, $\varepsilon > 0$, and $r \geq 1$ be an integer. We say that x is (ε, r) *so-pavable over* A if for every finite $F \subset H$ and every $\delta > 0$ there are a partition $P = (p_1, \dots, p_r)$ in A , an element $a = a^* \in A$ with $\|a\| \leq \|x\|$, and a projection $q \in M$ such that

$$\|(1 - q)\xi\| < \delta \quad (\xi \in F), \quad \|q(S_P(x) - a)q\| \leq \varepsilon \|x\|.$$

The integer r is fixed before F and δ are chosen.

Theorem 1.1. *For every $0 < \varepsilon < 1$, set*

$$r_\varepsilon = \left\lceil 400000000\varepsilon^{-2} \right\rceil + \lceil 4/\varepsilon \rceil + 1 \leq 500000000\varepsilon^{-2}.$$

For every masa $A \subseteq M \subseteq \mathcal{B}(H)$ and every $x = x^ \in M$, the element x is $(\varepsilon, r_\varepsilon)$ so-pavable over A . In particular, the number of projections is fixed before the inclusion, operator, and strong neighborhood. No separability or conditional-expectation assumption is imposed on the inclusion.*

Theorem 1.1 proves the quadratic so-paving conjecture of Popa and Vaes [PV15, Conjecture 2.8(2)]. Their definition of so-paving uses arbitrary strong neighborhoods of zero; Definition 1.1 gives the equivalent finite-vector formulation [PV15, Definition 2.2]. For self-adjoint x , allowing a general diagonal element in their definition gives the same condition, since taking its real part preserves the norm bound and the compression estimate.

For countably decomposable expected masas, the uniform bound also yields norm paving in the associated ultrapower.

Corollary 1.2 (Quadratic norm paving in Ocneanu ultrapowers). *Let $A \subseteq M$ be a masa. Suppose that A is countably decomposable and is the range of a normal conditional expectation from M . Let ω be a free ultrafilter on $\mathbb{N}$, and let $A^\omega \subseteq M^\omega$ be the associated inclusion in the Ocneanu von Neumann ultrapower. For every $0 < \varepsilon < 1$ and every $x = x^* \in M^\omega$, there are a partition $p_1, \dots, p_{r_\varepsilon} \in A^\omega$ and an element $a = a^* \in A^\omega$ such that*

$$\|a\| \leq \|x\|, \quad \left\| \sum_{i=1}^{r_\varepsilon} p_i x p_i - a \right\| \leq \varepsilon \|x\|.$$

Here r_ε is the same integer as in Theorem 1.1.

Proof. Theorem 1.1 makes every self-adjoint element of M $(\varepsilon, r_\varepsilon)$ so-pavable. The forward transfer in [PV15, Theorem 2.7(3)] gives the stated norm paving in M^ω with the same error and number of projections. Taking the real part of the diagonal element preserves both estimates. $\square$

1.1 History and significance

The classical Kadison–Singer paving problem concerns the diagonal masa in $\mathcal{B}(\ell^2\mathbb{N})$. Marcus, Spielman, and Srivastava resolved it by their mixed-characteristic-polynomial method [MSS15]. For general masas, norm paving is much more restrictive: when M has separable predual, it holds exactly when M is type I and A is the range of a normal conditional expectation [PV15, Theorem 3.3]. Strong-operator paving retains a uniform partition count while allowing a compression that approaches the identity strongly.

Popa’s II_1 -factor approach established paving for singular-masa ultraproducts [Pop14]. Popa and Vaes then obtained so-paving bounds of order ε^{-4} for arbitrary masas in type I algebras with separable predual, for Cartan masas in amenable algebras, and for the Cartan inclusions associated with essentially free ergodic probability-measure-preserving profinite actions of countable groups [PV15, Proposition 3.2 and Theorem 4.1]. In contrast, their theorem for tracial ultraproducts of singular masas in finite algebras gives a common norm-paving partition of order ε^{-2} for each finite family of centered self-adjoint contractions [PV15, Theorem 5.1]. Their later work gives the optimal uniform error for singular-masa ultraproducts [PV16, Theorem 4.1]. They explicitly identified the need to unite the Cartan and singular arguments [PV15, Remark 2.9(ii)]. On the finite-dimensional side, Ravichandran and Srivastava obtained asymptotically optimal multipaving bounds from mixed determinantal polynomials [RS17]; these supply the finite input used here.

A previous combination result illustrates the quantitative issue. For an intermediate inclusion $A \subseteq N \subseteq M$ with a faithful normal tracial state, Popa and Vaes assume that $L^2(M \ominus N)$ has no nonzero A – N subbimodule finitely generated on the right. Their bound uses at most $400\varepsilon^{-2}$ times the so-paving count for $A \subseteq N$ at error $\varepsilon/2$ [PV15, Remark 5.3]. Inserting their quartic Cartan bound gives order ε^{-6} . The present argument treats the normalizer algebra through a nonsingular measured equivalence relation and uses a single partition for both terms in the normalizer decomposition of one operator. This retains quadratic order. The finite-family obstruction in [PV15,

Remark 5.2] concerns the stronger requirement of a common partition for arbitrarily many separately prescribed operators in a finite ambient algebra.

The exponent two is forced already by the L^2 obstruction in a II_1 factor from [PV15, Proposition 5.4]. Fix an integer $1 \leq n < \varepsilon^{-2}$, choose $\varepsilon < \eta < n^{-1/2}$, and use the centered self-adjoint unitary x from that proposition at threshold η . Write τ for the normalized trace and $\|b\|_2 = \tau(b^*b)^{1/2}$. For every partition P with n pieces and every $a \in A$, trace orthogonality gives

$$\|S_P(x) - a\|_2^2 = \|S_P(x)\|_2^2 + \|a\|_2^2.$$

If $\|a\| \leq 1$, set $z = S_P(x) - a$, so $\|z\| \leq 2$. Whenever $\|qzq\| \leq \varepsilon$, the two discarded corner terms give

$$\|S_P(x)\|_2 \leq \|z\|_2 \leq \varepsilon + 4\tau(1 - q)^{1/2}.$$

In the trace representation, a strong neighborhood can force $\tau(1 - q) < ((\eta - \varepsilon)/4)^2$. The displayed estimate would then contradict the proposition's lower bound $\|S_P(x)\|_2 > \eta$, which holds for every such P . Thus its obstruction persists with the varying diagonals and compressions allowed here. The exponent in Theorem 1.1 is optimal, although its numerical constant is not optimized.

Theorem 1.1 controls a compression of the pinching of the original x . A separate expectation-free approximation-paving theorem instead supplies self-adjoint approximants y near x strongly, with $\|y\| \leq 3\|x\|$, and norm-paves them with error measured against $\|y\|$ [Ope26, Theorem 1.1]. Its arbitrary-masa conclusion has a finite ε -dependent count but states no quadratic rate. The two proofs are independent.

1.2 Outline of the proof

Let $p \in A$ be the supremum of the supports of all normal positive functionals ρ satisfying $\rho(bz) = \rho(zb)$ for $b \in A$ and $z \in M$. After the finite vector tests are fixed, we choose one such support $e \leq p$ so that $p - e$ is small on those vectors. When $e \neq 0$, the corner eMe carries a faithful normal state central over eAe . On $(1 - p)M(1 - p)$, a symmetric Haagerup L^2 estimate and two randomizations give a paving with $O(\varepsilon^{-1})$ colors; a positive-cone overlap estimate converts the symmetric estimate into a compression of arbitrarily large prescribed state mass. The corner $p - e$ is discarded. Since p and e need not be central in M , the construction uses disjoint palettes on these corners so that pinching removes cross-corner terms and the counts add. A separable reduction on the faithful-state corner then permits measured-space methods. These reductions are proved in Section 2.

In the presence of this central state, let N be the algebra generated by the partial normalizers of A . There are φ -preserving conditional expectations $E_A : M \rightarrow A$ and $E_N : M \rightarrow N$, and $A \subseteq N$ is Cartan. We use the decomposition

$$x - E_A x = (E_N x - E_A x) + (x - E_N x).$$

Section 3 proves a structural fact about the second term: for $y \in \ker E_N$, the positive map $f \mapsto E_A(y^*fy)$ on A is represented by an atomless kernel. This is the input needed to remove coincidences in the later moment transfer.

The Cartan term is treated on a relative Bernoulli extension of the measured equivalence relation representing $A \subseteq N$. Each relation class receives independent site labels. We construct a measurable coloring there that has uniform conditional color marginals and makes the pinched matrix small on any prescribed finite-radius ball with arbitrarily high probability. This is Theorem 6.1. Its proof occupies Sections 4–6.

The finite-dimensional input to this construction is the mixed determinantal bound of [RS17, Theorem 25]. For a partial coloring, a real-rooted polynomial uses both signs of the matrix and

shifts determined by the assigned sites; the sum of its positive roots is the total cost. At the chosen shift, the upper mixed bound makes the all-unassigned cost zero, while the lower comparison for the two signs makes excessive norm on a fully assigned finite set force positive cost. Rooted spectral measures allocate this cost to vertices, and a single-site assignment inequality controls its increase. For a nonsingular relation, an invariant lift with an extra real coordinate makes mass transport available. A deletion estimate controls large weights uniformly, so joint cutoff limits give homogeneous costs on the original probability space. Random assignment times produce a partial coloring of arbitrarily small total cost for which any prescribed ball is fully assigned with high probability. Completing the remaining sites preserves the estimate on those balls. The uniform cutoff estimates and the order of conditioning on the assignment times are part of the construction.

To combine the two operator terms, we place M and the Bernoulli extension of N in their amalgamated free product over N . Uniform conditional color marginals give a free compression bound on $\ker E_N$, while the same color projections provide the Cartan estimate. Section 7 transfers each fixed mixed scalar moment from this model to actual projections of A . Its proof uses finite-symbol evaluation, modular analytic approximation, and a shortest-pair argument in cyclic words. The atomless kernels make the remaining collision terms vanish.

Finally, Section 8 converts the local Cartan estimates to bounded approximants by clipping, controls continuous calculus for the resulting varying sequences, and transfers a sufficiently high fixed moment. A spectral cut produces the required projection q . All approximation parameters are chosen after the number of colors, and the corner assembly completes Theorem 1.1.

1.3 Conventions

We allow zero projections in a partition and use 1 also for the identity of a corner when that corner is regarded as an algebra. Except when restoring scale in the final proof, the operators to be paved are self-adjoint contractions. Maximal abelianness implies that A is weakly closed and that $A' \cap M = A$: its weak closure is abelian, and any self-adjoint element of the relative commutant could otherwise be adjoined. In particular, corners by projections in A are again masas in the corresponding corners of M .

For a normal positive functional φ , write $\|z\|_\varphi = \varphi(z^*z)^{1/2}$. When φ is a faithful normal state, its centralizer is the algebra of elements fixed by its modular automorphism group. Measurable statements are understood modulo null sets. Whenever a countable nonsingular relation is present, a null exceptional set can be enlarged to its null saturation.

2 Reduction to a faithful central state

We first remove the obstruction to using a normal conditional expectation: the original masa need not be the range of one. A normal positive functional ρ on M is *A-central* if

$$\rho(bz) = \rho(zb) \quad (b \in A, z \in M).$$

The supremum p of their support projections lies in A , as proved in the corner assembly below. Given finite vector tests, that assembly chooses one support $e \leq p$ so that $p - e$ acts arbitrarily small on those vectors. It uses a faithful central state on eMe , a direct randomization on $(1 - p)M(1 - p)$, and discards $p - e$. The projections p and e need not be central in M ; disjoint palettes make the pinching remove all cross-corner terms, so the numbers of colors add. We then reduce the faithful-state case to separable predual. All these reductions retain a bound on the number of projections that is independent of the prescribed strong neighborhood.

2.1 The part without central normal functionals

For the first reduction we use Haagerup's L^p spaces for an arbitrary von Neumann algebra, constructed with a normal semifinite faithful weight; see [Haa79, Section 1]. If φ is a normal positive functional, its density $h_\varphi \in L^1(M)_+$ represents it under the isometric identification $L^1(M) = M_*$. The pairing is denoted by Tr . The space $L^2(M)$, with left and right multiplication and its positive cone, is a standard form of M . In particular its cone vector for φ is $h_\varphi^{1/2}$. We use Hölder's inequality, cyclicity of the L^p – $L^{p'}$ pairing, and polar decomposition, whose partial isometry belongs to M .

The positive-cone estimate below is Haagerup's standard-form inequality [Haa75, Lemma 2.10]; see also the earlier work of Powers and Størmer [PS70, Lemma 4.1] and Araki [Ara74, Theorem 4(8), equation (5.10), and Section 8, Remark 2]. We include its proof so that no faithfulness assumption on the two functionals is implicit. For normal positive φ, ψ ,

$$\|h_\varphi^{1/2} - h_\psi^{1/2}\|_2^2 \leq \|\varphi - \psi\|. \quad (2.1)$$

Indeed, put $\xi_1 = h_\varphi^{1/2}$, $\xi_2 = h_\psi^{1/2}$ and $D = \xi_1 - \xi_2$. Its sign $s \in M$ satisfies $\|s\| \leq 1$ and $sD = Ds = |D|$. Writing $D = D_+ - D_-$, cyclicity gives

$$\begin{aligned} \text{Tr}(s(h_\varphi - h_\psi)) &= \text{Tr}(s(D\xi_1 + \xi_2 D)) = \text{Tr}(|D|(\xi_1 + \xi_2)) \\ &= \|D\|_2^2 + 2\text{Tr}(D_- \xi_1) + 2\text{Tr}(D_+ \xi_2) \geq \|D\|_2^2. \end{aligned}$$

The last two pairings are nonnegative by self-duality of the positive cone, and the first expression is at most $\|\varphi - \psi\|$.

Lemma 2.1. *Let $A \subseteq M$ be a masa such that M has no nonzero A -central normal positive functional. Let $x = x^* \in M$ satisfy $\|x\| \leq 1$, and let $0 < \varepsilon < 1$. Put $r_p = \lceil 4/\varepsilon \rceil$. For every normal state φ on M and every $\rho > 0$ there are a partition $p_1, \dots, p_{r_p}$ in A , an $a = a^* \in A$ with $\|a\| \leq 1$, and a projection $q \in M$ such that*

$$\varphi(1 - q) < \rho, \quad \left\| q \left(\sum_{i=1}^{r_p} p_i x p_i - a \right) q \right\| \leq \varepsilon.$$

Proof. Write $\xi = h_\varphi^{1/2}$ and introduce the symmetric map

$$T(z) = h_\varphi^{1/4} z h_\varphi^{1/4} : M \longrightarrow L^2(M).$$

It is bounded and takes weak-star convergence to weak convergence: testing against $\zeta \in L^2(M)$ uses the L^1 element $h_\varphi^{1/4} \zeta^* h_\varphi^{1/4}$. For a finite partition P in A , the pinching $S_P = \sum_{e \in P} e(\cdot)e$ also acts on $L^2(M)$. There it is an orthogonal projection. These projections decrease under refinement to the projection onto the A -central vectors. Such a vector must be zero: otherwise its left vector functional would be a nonzero A -central normal positive functional. To see centrality, move a unitary of A from the left of the vector to its right, where it commutes with the left action of M . Consequently

$$\|S_P(\xi)\|_2 \longrightarrow 0.$$

For $z \in M$, cyclicity and Hölder's inequality give

$$\sum_{e, f \in P} \|T(ezf)\|_2^2 = \text{Tr}(z^* S_P(\xi) z S_P(\xi)) \leq \|z\|^2 \|S_P(\xi)\|_2^2. \quad (2.2)$$

In particular $\|T(eze)\|_2^2 \leq \|e\xi e\|_2^2$ when $\|z\| \leq 1$.

Fix $\kappa > 0$ and a coarse partition P_0 with $\|S_{P_0}(\xi)\|_2 < \kappa$. A weak-star convergent subnet of $S_P(x)$, as P refines P_0 , has a self-adjoint contraction limit $b \in A$: the limit commutes with every partition in A , hence belongs to the masa. Since the weak and norm closures of a convex subset of a Hilbert space agree, there are refinements $P^1, \dots, P^d$ of P_0 and probabilities $t_1, \dots, t_d$ such that

$$\left\| \sum_{j=1}^d t_j T(S_{P^j}(x)) - T(b) \right\|_2 < \kappa.$$

For each $e \in P_0$, independently choose j_e with this distribution and use the partition P^{j_e} inside e . The resulting partition R refines P_0 . The mean of $T(S_R(x))$ is the displayed convex combination. Independence and Equation (2.2) bound its variance by

$$\sum_{e \in P_0} \|e\xi e\|_2^2 = \|S_{P_0}(\xi)\|_2^2 < \kappa^2.$$

Thus one realization satisfies $\|T(S_R(x) - b)\|_2 < \sqrt{2}\kappa$.

Now color the pieces of this fixed R independently and uniformly with r_p colors, and let S_{col} be the resulting color pinching. Its mean is

$$\mathbb{E}S_{\text{col}}(x) = \frac{x}{r_p} + \left(1 - \frac{1}{r_p}\right) S_R(x).$$

The indicators of equal colors for distinct unordered pairs of pieces are uncorrelated, even when the pairs share a piece. Applying $\|u + v\|_2^2 \leq 2\|u\|_2^2 + 2\|v\|_2^2$ and Equation (2.2), the variance after applying T is at most $2\|S_R(\xi)\|_2^2 < 2\kappa^2$. Therefore a coloring can be chosen so that the self-adjoint element

$$y = S_{\text{col}}(x) - \left(1 - \frac{1}{r_p}\right) b - \frac{x}{r_p}$$

satisfies $\|T(y)\|_2 < 2\sqrt{2}\kappa$. Also $\|y\| \leq 2$. Since κ is arbitrary, this gives arbitrarily small symmetric seminorm with the number of colors already fixed.

It remains to convert this seminorm into a large compression. Set $t = \|T(y)\|_2$ and $\psi(z) = \varphi(y^*zy)$. Small t need not control $\|y\|_\varphi$, so we first seek q_0 with both $\varphi(1 - q_0)$ and $\psi(q_0)$ small; these bounds will control $\|q_0 y q_0\|_\varphi$ for the spectral cut. Polar decomposition gives $h_\psi = y h_\varphi y^*$ and $h_\psi^{1/2} = y \xi u^*$ for a partial isometry $u \in M$. Hence

$$0 \leq \text{Tr}(\xi h_\psi^{1/2}) \leq \|\xi y \xi\|_1 = \|h_\varphi^{1/4} T(y) h_\varphi^{1/4}\|_1 \leq t.$$

Here the first inequality uses the positive cone, and the last uses $\|h_\varphi^{1/4}\|_4 = 1$. Let q_0 support the positive part in the Jordan decomposition of $\varphi - \psi$. Equation (2.1) implies

$$\varphi(1 - q_0) + \psi(q_0) = \frac{\varphi(1) + \psi(1) - \|\varphi - \psi\|}{2} \leq t. \quad (2.3)$$

Moreover

$$\|q_0 y q_0 \xi\|_2 \leq \|q_0 y \xi\|_2 + \|q_0 y (1 - q_0) \xi\|_2 \leq \psi(q_0)^{1/2} + 2\varphi(1 - q_0)^{1/2} \leq 3\sqrt{t}.$$

Inside $q_0 M q_0$, let q be the spectral projection of $q_0 y q_0$ for $[-\varepsilon/2, \varepsilon/2]$. The spectral Markov inequality gives

$$\varphi(1 - q) \leq \varphi(1 - q_0) + \frac{4}{\varepsilon^2} \varphi((q_0 y q_0)^2) \leq \left(1 + \frac{36}{\varepsilon^2}\right) t. \quad (2.4)$$

Taking κ sufficiently small makes this less than ρ . With $a = (1 - 1/r_p)b$, we have $\|a\| \leq 1$ and

$$\|q(S_{\text{col}}(x) - a)q\| \leq \frac{\varepsilon}{2} + \frac{1}{r_p} \leq \frac{3\varepsilon}{4}.$$

This proves the lemma. $\square$

2.2 Assembly of corners

Proposition 2.2. *Fix $0 < \varepsilon < 1$ and an integer $R \geq 1$. Suppose that for every masa $A \subseteq M$ admitting a faithful A -central normal state φ , every self-adjoint contraction $x \in M$, and every $\rho > 0$, there exist a partition of 1 into R projections of A , a self-adjoint $a \in A$ of norm at most one, and a projection $q \in M$ with $\varphi(1 - q) < \rho$ and*

$$\left\| q \left(\sum_{i=1}^R p_i x p_i - a \right) q \right\| \leq \varepsilon.$$

Then every masa in every represented von Neumann algebra has the strong-operator paving property of Theorem 1.1 at error ε , with

$$r = R + \lceil 4/\varepsilon \rceil + 1.$$

In particular this number is chosen before the finite vector set and the strong-neighborhood tolerance.

Proof. Let p be the supremum of the supports of the A -central normal positive functionals on M . Each support belongs to A , since conjugation by every unitary of A preserves the functional and its support. These supports form a directed family: the support of a sum is the join of the supports. The corner $(1 - p)M(1 - p)$ has no nonzero $A(1 - p)$ -central normal positive functional, since compression would extend one to an A -central functional on M .

Given a finite $F \subset H$ and $\delta > 0$, set

$$\chi(z) = \sum_{\xi \in F} \langle z\xi, \xi \rangle.$$

Normality supplies a support $e \leq p$ in the directed family with $\chi(p - e)$ as small as desired. On the nonzero corner eMe , the corresponding functional restricts, after normalization, to a faithful normal state central on Ae .

For any faithful normal state ω , small ω -mass on projections implies uniformly small mass for each fixed normal positive functional. Indeed, a contrary sequence in the positive unit ball would have a weak-star convergent subnet whose limit has zero ω -mass but positive mass for that functional. Faithfulness forces the positive limit to be zero, a contradiction. Apply the assumed result on eMe , using this observation to make the lost χ -mass small. On $(1 - p)M(1 - p)$ apply Lemma 2.1 to the normalized restriction of χ if it is nonzero; otherwise take $q = 0$ on this corner. Empty corners require no construction.

Use disjoint palettes for the two corner partitions, and the additional projection $p - e$ as the last piece. Take corner sums of the two diagonal terms and of the two large projections, with zero on $p - e$. Every pinching piece is supported in one of the three corners, so all off-corner terms of x disappear. Thus the compressed operator norm is at most ε , and the diagonal norm is at most one. The choices can ensure $\chi(1 - q) < \delta^2$, which gives

$$\|(1 - q)\xi\|^2 \leq \chi(1 - q) < \delta^2 \quad (\xi \in F).$$

Zero projections pad either palette when necessary. The total number is the stated r , independently of F and δ . Scaling handles an arbitrary nonzero self-adjoint x ; for $x = 0$ the assertion is immediate. $\square$

2.3 State norm and separable reduction

We now work with a faithful A -central normal state φ . Write

$$\|z\|_\varphi = \varphi(z^*z)^{1/2}, \quad M_\varphi = \{b \in M : \varphi(bz) = \varphi(zb) \text{ for all } z \in M\}.$$

The centralizer M_φ is the fixed algebra of the modular group σ^φ . In particular $A \subseteq M_\varphi$. We first obtain approximation by diagonal pinchings in the state norm, then use it to construct a separable inclusion containing the operator.

Lemma 2.3. *Let $A \subseteq M$ be a masa and φ a faithful A -central normal state. There is a faithful normal φ -preserving conditional expectation $E_A: M \rightarrow A$. On bounded subsets of M , convergence to zero in $\|\cdot\|_\varphi$ is equivalent to strong convergence in the faithful GNS representation. For every $z \in M$,*

$$\|S_P(z) - E_A(z)\|_\varphi \rightarrow 0 \tag{2.5}$$

as the finite partitions P of A refine. The same conclusion holds along any increasing sequence of finite partitions generating A .

Proof. Takesaki's Conditional Expectation Theorem [Tak72] applies because the modular group fixes A pointwise; both the state and its restriction are faithful, normal, and finite. Let Ω be the GNS state vector. It is separating and therefore cyclic for the commutant. If a bounded net z_j satisfies $z_j\Omega \rightarrow 0$, then $z_j c' \Omega = c' z_j \Omega \rightarrow 0$ for every commutant element c' . Density and boundedness give strong convergence. The converse follows by testing on Ω .

Since the partition projections lie in the centralizer, S_P induces an orthogonal projection on the GNS space; equivalently, it is the state-preserving conditional expectation onto the commutant in M of the projections of P . These Hilbert-space projections decrease under refinement. On the vector $z\Omega$, let their limit be η . A bounded weak-star cluster point b of $S_P(z)$ commutes with every partition of A and hence lies in A . Its state vector is η . Moreover $E_A S_P(z) = E_A(z)$, so $b = E_A(z)$. This proves Equation (2.5). If an increasing sequence generates A , its limit commutes with the generating projections, and the same argument applies. $\square$

Lemma 2.4. *Let $A \subseteq M$ be a masa with a faithful A -central normal state φ , and let $x = x^* \in M$. There exist unital von Neumann subalgebras $A_0 \subseteq A$ and $M_0 \subseteq M$ such that $x \in M_0$, M_0 has separable predual, A_0 is a masa in M_0 , and $E_A(M_0) \subseteq A_0$. The restriction of φ is faithful, normal, and A_0 -central. Thus a uniform faithful-state paving result for inclusions with separable predual implies the same result without that restriction.*

Proof. Start with $D_1 = W^*(1, x)$. Inductively, choose a countable $\|\cdot\|_\varphi$ -dense subset $\mathcal{D}_j$ of the unit ball of D_j . Such a set exists for every countably generated D_j : the separable C^* -algebra of its generators has dense state vectors by Kaplansky's Density Theorem. Its GNS representation is faithful and normal on a separable Hilbert space, so D_j has separable predual. Let D_{j+1} be generated by D_j , all $E_A(z)$ for $z \in \mathcal{D}_j$, and the projections of finite partitions $P(z, k) \subset A$ satisfying

$$\|S_{P(z,k)}(z) - E_A(z)\|_\varphi < 1/k \quad (z \in \mathcal{D}_j, k \geq 1).$$

Lemma 2.3 supplies these partitions. Only countably many generators have been added.

Put $M_0 = W^*(\bigcup_j D_j)$ and let A_0 be generated unilaterally by all the adjoined elements of A . Bounded Kaplansky approximation by the union algebra, followed by approximation from the sets $\mathcal{D}_j$, shows that their union is state-norm dense in the unit ball of M_0 . Here one may first approximate in the norm closure of the union and then make a norm approximation and a harmless

rescaling to preserve the bound. Since E_A is state-norm contractive, every $E_A(z)$ for $z \in M_0$ is a bounded state-norm limit of elements of A_0 . Lemma 2.3 and strong closedness imply $E_A(z) \in A_0$.

If a unit-ball element $z \in M_0$ commutes with A_0 , each selected partition satisfies $S_P(z) = z$. For $z' \in \mathcal{D}_j$ and its selected partition $P = P(z', k)$, contractivity gives

$$\|z - E_A(z)\|_\varphi \leq 2\|z - z'\|_\varphi + \|S_P(z') - E_A(z')\|_\varphi.$$

The right side can be made arbitrarily small; faithfulness gives $z = E_A(z) \in A_0$. Thus A_0 is maximal abelian in M_0 . The algebra M_0 is countably generated and has separable predual by the same GNS argument. All the assertions about the restricted state are immediate. Any paving produced in M_0 has the same operator-norm bound and the same lost φ -mass when regarded in M . This is the faithful-state estimate required by Proposition 2.2. $\square$

3 The normalizing algebra and atomless kernels

For the rest of the structural argument, M has separable predual, $A \subseteq M$ is a masa, and φ is a faithful A -central normal state. Lemma 2.4 permits these assumptions. We separate the algebra generated by normalizers from its orthogonal complement with respect to a state-preserving expectation. The former has a measured-relation description. On the latter, we prove that the positive maps $f \mapsto E_A(y^*fy)$ have atomless kernels. This atomlessness will make fine-partition collision terms vanish in the moment comparison.

3.1 Normalizers and the measured relation

A *groupoid normalizer* of A is a partial isometry $v \in M$ such that $v^*v, vv^* \in A$ and

$$vAv^* = Avv^*, \quad v^*Av = Av^*v.$$

Let N be the von Neumann algebra they generate. For every such v , the modular transform $\sigma_t^\varphi(v)$ induces the same isomorphism between the two corners of A , since σ_t^φ fixes A pointwise. Consequently $v^*\sigma_t^\varphi(v)$ is a unitary in Av^*v . These unitaries form a strongly continuous one-parameter group: the cocycle identity becomes the group identity because the modular group fixes Av^*v . Thus there is a positive nonsingular operator h_v affiliated with Av^*v such that

$$\sigma_t^\varphi(v) = vh_v^{it} \quad (t \in \mathbb{R}). \quad (3.1)$$

In particular N is invariant under σ^φ . Takesaki's Theorem [Tak72] gives a faithful normal φ -preserving conditional expectation

$$E_N: M \longrightarrow N, \quad M^\circ = \ker E_N.$$

The normalizers $v1_{[-k,k]}(\log h_v)$ are entire analytic for σ^φ and converge strongly-* to v . Indeed Equation (3.1) extends on this cut to $vh_v^{iz}1_{[-k,k]}(\log h_v)$ for $z \in \mathbb{C}$, a norm-entire function.

Identify $A = L^\infty(X, \mu)$ on a standard probability space, with $\varphi|_A$ given by integration. A groupoid normalizer induces a measure-class preserving isomorphism between measurable subsets of X . Its restriction to the fixed points belongs to A . The remaining domain can be partitioned into countably many measurable pieces each disjoint from its image: use a countable family separating distinct points to obtain a countable cover by such pieces, and then disjointize the cover. On one piece let the cut normalizer be v' . Its initial and final projections are orthogonal, and

$$v' + v'^* + (1 - v'^*v' - v'v'^*)$$

is a unitary normalizer. Recovering v' by a corner cut and summing strongly shows that full unitary normalizers also generate N . The pair $A \subseteq N$ is therefore Cartan: A is maximal abelian, regular, and the range of the faithful normal expectation $E_A|_N$.

The Feldman–Moore Representation Theorem [FM77, Theorem 1] realizes this pair as the algebra of a countable nonsingular measured equivalence relation $\mathcal{R}$ on (X, μ) , possibly twisted by a scalar 2-cocycle. The separably acting hypothesis holds in the faithful state GNS representation. We use the right counting representation

$$L^2(\mathcal{R}, \nu_r) = \int_X^\oplus \ell^2([v]_{\mathcal{R}}) d\mu(v), \quad \int f d\nu_r = \int_X \sum_{u \sim v} f(u, v) d\mu(v). \quad (3.2)$$

Here A acts diagonally by the function of the left site. With a normalized cocycle s , a kernel a acts in the fiber rooted at v by the matrix

$$(a(u, w)s(u, w, v))_{u, w \in [v]_{\mathcal{R}}}.$$

The identity

$$s(u, w, v)s(u, v, v') = s(u, w, v')s(w, v, v')$$

shows that rerooting identifies these matrices by diagonal unitary conjugacy. Thus their norms and their conjugacy-invariant principal matrix data do not depend on the chosen root.

Partial isomorphisms with graphs in $\mathcal{R}$ lift to kernel groupoid normalizers. Their finite sums with bounded base coefficients form a weakly dense $*$ -algebra, denoted N_{alg} ; see [FM77, Section 2]. The canonical expectation to A takes the diagonal. It agrees with $E_A|_N$: on the fixed part of a normalizer both maps retain its diagonal value, and on pieces with disjoint initial and final supports both vanish by A -bimodularity. The identity extends by normality. Therefore $\varphi|_N$ is diagonal integration, represented in Equation (3.2) by the unit site vectors δ_v . All data are understood modulo null sets, with Borel versions on standard spaces. Countably many null exceptions may be discarded along with their relation saturation, which remains null by nonsingularity.

3.2 A type I observation

The kernel proof will produce an intertwiner whose initial projection is not yet known to belong to A . The following observation is the step that turns this intertwiner into a groupoid normalizer. Its state hypothesis excludes the diffuse masas that can occur in a type I algebra without a normal conditional expectation. The importance of that expectation hypothesis is also reflected in the type I criterion of [AS12, Theorem 4.1]; we prove the precise normalizer and abelian-projection consequences needed here.

Lemma 3.1. *Let D be a type I von Neumann algebra with separable predual, and let $C \subseteq D$ be a masa. Suppose D admits a faithful C -central normal state. Then the groupoid normalizers of C generate D . Every abelian projection of D is Murray–von Neumann equivalent in D to a projection of C .*

Proof. Use the standard direct integral decomposition of a type I algebra over its center into full operator algebras; see [KR86, Sections 6.6 and 14.1–14.2]. Integrating ordinary fiber traces against a probability measure in the center measure class gives a faithful normal semifinite trace τ . The given state has a positive trace density h of full support, with $\tau(h) = 1$; this is the predual identification in the semifinite case [Haa79, Proposition 1.5 and Theorem 2.1]. Every unitary of C preserves

the state and hence its density. Thus h commutes with C in the affiliated-operator sense, and its spectral projections belong to C by maximal abelianness. The increasing projections

$$f_n = 1_{[1/n, \infty)}(h), \quad n \geq 1,$$

exhaust 1 and satisfy $\tau(f_n) \leq n$. Their successive differences form a countable partition of 1 in C into trace-finite projections. Such a projection has finite rank almost everywhere in the type I fibers. Split it further by central projections according to its rank. Each resulting corner is a measurable field of fixed finite matrix size, with its compressed C a masa.

We justify explicitly the measurable diagonalization needed in these finite corners. Trivialize the finite-rank field and choose countably many commuting self-adjoint measurable matrix fields generating the compressed C . In each fiber, the algebras generated by the first j fields eventually equal the algebra generated by the whole list, since their dimensions are nondecreasing bounded integers. The first index attaining the limiting dimension is measurable: each finite dimension is determined by ranks of finite lists of polynomial matrices, and the limiting dimension is their countable supremum. Split the center according to this first index. On each piece, successive eigenspace decompositions measurably diagonalize the finite commuting list. Ordered eigenvalues give measurable spectral projections, and Gram–Schmidt applied to fixed coordinate vectors gives measurable orthonormal bases of their ranges. The resulting rank-one projection fields commute with all generators, not just the chosen initial list. As bounded elements of the ambient field algebra, they therefore belong to the masa.

We have obtained countably many rank-one fields $e_j \in C$, allowing zero outside their central domains, with $\sum_j e_j = 1$. On $e_j D e_j$ the algebra $C e_j$ consists precisely of the center scalars over that domain, since the center of D is contained in C . Measurable matrix units between e_j and e_k on their common central domain therefore partially normalize C . These matrix units and center coefficients generate D : compress a bounded field first by finite sums of the e_j , express its matrix entries using these units, and pass to the strong limit.

Finally, an abelian projection $e \in D$ has rank one almost everywhere on its central support z . Choose, fiber by fiber on z , the first nonzero e_j from the countable decomposition. This gives a projection $e' \in C$ with the same central support and rank. Measurable unit vectors spanning the ranges of e and e' give a partial isometry $w \in D$ with $ww^* = e$ and $w^*w = e'$. The assertion also holds for $e = 0$. $\square$

3.3 Atomless kernels off the normalizing algebra

A positive kernel on X means a measurable family of finite positive Borel measures $K(v, du)$. It represents a positive normal map $K: A \rightarrow A$ if

$$(Kf)(v) = \int_X f(u) K(v, du) \quad (f \in L^\infty(X, \mu)),$$

where for each fixed μ -null Borel set S , $K(v, S) = 0$ for almost every v . This condition makes the formula well defined on equivalence classes; it does not assert that every row measure is absolutely continuous with respect to μ .

Proposition 3.2. *In the setting of this section, for every $y \in M^\circ$ the map*

$$K_y(f) = E_A(y^* f y), \quad f \in A,$$

has a positive kernel whose mass is at most $\|y\|^2$ and which is atomless for almost every output point $v \in X$. For $k \geq 1$ and $y_1, \dots, y_k \in M^\circ$, the composition $K_{y_1} \circ \dots \circ K_{y_k}$ has a positive kernel whose

mass is at most $\prod_{j=1}^k \|y_j\|^2$ and which is atomless almost everywhere. If $p_{\text{at}} \in A$ is the projection of the atomic part of A , then

$$p_{\text{at}}y = yp_{\text{at}} = 0 \quad (y \in M^\circ).$$

Proof. Let Ω be the standard state vector for φ . The commuting left and right actions of A have a joint spectral measure on $X \times X$ on the vector $y\Omega$. Since $A \subseteq M_\varphi$, the right action of $f \in A$ on this vector is $yf\Omega$. Thus the joint vector measure λ satisfies

$$\lambda(S \times T) = \varphi(y^*1_S y 1_T) = \int_T E_A(y^*1_S y) d\mu \quad (3.3)$$

for Borel $S, T \subseteq X$. Disintegrating relative to the right coordinate gives a kernel for K_y with mass $E_A(y^*y)(v) \leq \|y\|^2$. Choose a Borel version and set its rows to zero on a Borel null exceptional set so that this mass bound holds for every row. For every fixed μ -null S , Equation (3.3) gives $K_y(v, S) = 0$ almost everywhere. This construction uses the joint spectral measure and does not require product-measure absolute continuity.

Suppose this kernel is not almost everywhere atomless. The function $(v, u) \mapsto K_y(v, \{u\})$ is measurable: express it as the decreasing limit of the masses of the pieces containing u in an increasing sequence of finite Borel partitions that separates points. For some integer $k \geq 1$, the Borel set where this function is at least $1/k$ has finite vertical sections and a projection of positive measure. The Lusin–Novikov Theorem [Kec95, Theorem 18.10] supplies a measurable selection $s: T \rightarrow X$ on a positive-measure Borel set T such that

$$K_y(v, \{s(v)\}) \geq 1/k \quad (v \in T). \quad (3.4)$$

For a fixed μ -null Borel set S , this implies

$$\mu(\{v \in T : s(v) \in S\}) \leq k \int_T K_y(v, S) d\mu(v) = 0.$$

Consequently

$$\theta(f) = 1_T(f \circ s): A \longrightarrow A1_T$$

is a well-defined unital $*$ -homomorphism into the corner. It is normal: the pushforward under s of any finite measure with an L^1 density on T is absolutely continuous with respect to μ , and its Radon–Nikodym density gives the preadjoint of θ .

Choose increasing finite Borel partitions P_j of X which generate A and separate points, and put

$$b_j = \sum_{e \in P_j} ey\theta(e). \quad (3.5)$$

The left support projections and the right support projections are orthogonal families, so $\|b_j\| \leq \|y\|$. Also $E_N(b_j) = 0$ by N -bimodularity of E_N . Their state vectors are the decreasing joint projection cuts of $y\Omega$ to the sets where the left coordinate and s of the right coordinate lie in the same piece of P_j , with the right coordinate in T . Their limit η therefore satisfies

$$\|\eta\|^2 = \int_T K_y(v, \{s(v)\}) d\mu(v) > 0.$$

A weak-star convergent subnet of the bounded b_j has a limit $b \in M^\circ$ with $b\Omega = \eta$. Indeed testing the vectors against $z\Omega$, $z \in M$, gives the normal functionals $\varphi(z^*b_j)$, so their vector and weak-star limits agree. In particular $b \neq 0$. We have $b = b1_T$ and

$$fb = b\theta(f) \quad (f \in A). \quad (3.6)$$

For partition step functions this follows from refinement in Equation (3.5); bounded strong approximation and normality of θ extend it to all $f \in A$.

We show that Equation (3.6) forces $b \in N$. Write its polar decomposition as $b = u|b|$. The intertwining relations imply

$$|b^*| \in A, \quad fu = u\theta(f) \quad (f \in A), \quad e := u^*u \in D := \theta(A)' \cap 1_T M 1_T.$$

Indeed bb^* commutes with A , while b^*b commutes with $\theta(A)$, and polar decomposition preserves the resulting intertwining relation. For $d \in eDe$, the element udu^* commutes with A and belongs to M , so lies in A . It follows that e is an abelian projection in D . Let z be its central support in D . The abelian-projection characterization of type I algebras [KR86, Sections 6.4–6.6] shows that Dz is type I: every nonzero central part of Dz meets e and hence contains a nonzero abelian projection. Moreover $A1_T$ is a masa in D , so its center, and in particular z , belongs to $A1_T$.

The restricted state on Dz , after normalization, is faithful, normal, and central on the masa Az . By Lemma 3.1, the algebra Dz is generated by groupoid normalizers of Az . These also normalize A partially: their supports lie in Az , and every element of A restricts on z to an element of Az . Therefore $Dz \subseteq N$. The same lemma supplies $w \in Dz$ and $e' \in Az$ with

$$ww^* = e, \quad w^*w = e'.$$

Set $v = uw$. Its range projection $uu^* = \text{supp } |b^*|$ and its initial projection e' belong to A . Since w commutes with $\theta(A)$,

$$v^*Av = w^*u^*Auw = \theta(A)e' \subseteq Ae'.$$

The left side is a masa in $e'Me'$, by conjugating the masa on the range corner with v . Hence the inclusion is equality, and v is a groupoid normalizer of A . We conclude

$$u = vw^* \in N, \quad b = |b^*|u \in N,$$

contrary to $0 \neq b \in \ker E_N$. The kernel is therefore atomless almost everywhere.

For the composition assertion, let K_1, K_2 be two of the kernels. Choose one fixed μ -null Borel set S_0 outside which every row of K_2 is atomless. For almost every v , $K_1(v, S_0) = 0$. For each such v the measure

$$(K_1 \circ K_2)(v, E) = \int_X K_2(u, E) K_1(v, du)$$

vanishes on every singleton E ; the conclusion holds for all singletons simultaneously because the integrand row measures are atomless off S_0 . The mass bound and the fixed-null-set property also pass to this composition. Iteration proves the assertion for any finite composition.

Finally the atomic part of the standard probability space can be represented by a countable set of positive-mass points. Atomlessness gives $E_A(y^*p_{\text{at}}y) = 0$; faithfulness then gives $p_{\text{at}}y = 0$. Applying the same argument to $y^* \in M^\circ$ gives $yp_{\text{at}} = 0$. $\square$

4 Finite polynomials and the cost of assigning a color

We develop a finite-dimensional estimate for assigning colors to the sites of a Hermitian matrix. The polynomial associated with a partial coloring will average over the remaining color choices and over independent signs. We use the sum of the positive roots after a negative diagonal shift as a cost. Assigning a color restricts the average and makes the shift more negative at the chosen site; we will show that some color increases this cost by at most a fixed multiple of a quantity attached to that site. The locality and uniform bounds proved alongside this estimate will permit its use on measured graphs.

4.1 Finite data and mixed determinantal bounds

Fix a finite set I , a zero-diagonal Hermitian matrix $T = (T_{uv})_{u,v \in I}$ with $\|T\| \leq 1$, an integer $r \geq 1$, and $t > 0$. Throughout the coloring argument we use

$$\beta = 80.$$

A partial coloring is a function $c : I \rightarrow \{0, 1, \dots, r\}$, where 0 means unassigned. Give each site a weight $W_v > 0$, and put

$$\tau_v = \begin{cases} t, & c(v) = 0, \\ \beta t, & c(v) > 0, \end{cases} \quad a_v = \tau_v W_v.$$

At site v , the allowed pairs of color and sign are

$$\mathcal{A}_v = \begin{cases} \{1, \dots, r\} \times \{+1, -1\}, & c(v) = 0, \\ \{c(v)\} \times \{+1, -1\}, & c(v) > 0. \end{cases}$$

Choose a pair ω_v uniformly from $\mathcal{A}_v$, independently over sites, and write $k_v = |\mathcal{A}_v|$. For $j = (i, \sigma) \in \{1, \dots, r\} \times \{+1, -1\}$, set

$$U_j = \{v \in I : j \in \mathcal{A}_v\}, \quad B_j = \sigma D_W T D_W - \text{diag}(a_v)_{v \in I}, \quad D_W = \text{diag}(\sqrt{W_v})_{v \in I}.$$

For a realization ω , let $I_j(\omega) = \{v : \omega_v = j\}$. Its outcome matrix is the direct sum of the principal matrices $B_j|_{I_j(\omega)}$, placed on these subsets of I ; its entries between different subsets are zero. Denote this matrix by B_ω . With $Z = \text{diag}(z_v)_{v \in I}$, define

$$P_I(Z) = \mathbb{E}_\omega \det(Z - B_\omega), \quad P_I(z) = P_I(z, \dots, z). \quad (4.1)$$

We write $P = P_I$ when I is fixed. Restriction to a subset always means principal restriction of all the finite data, with the same laws at the remaining sites. In particular $P_{-v} = P_{I \setminus \{v\}}$, and $P_\emptyset = 1$. These conventions also apply to all polynomials introduced below.

We shall use the following two bounds from mixed determinantal polynomial theory. We state their precise normalization because both signs of T will enter our application.

Lemma 4.1 (Mixed determinantal bounds). *For a k -tuple of Hermitian matrices $A_1, \dots, A_k$ on a common nonempty finite set, let*

$$\chi[A_1, \dots, A_k](z) = \mathbb{E} \prod_{j=1}^k \det(zI - A_j|_{S_j}),$$

where each site is independently assigned uniformly to one of the k sets S_j , and the determinant of an empty matrix is 1. This polynomial is real-rooted. If $k \geq 2$ and the A_j are zero-diagonal contractions, then

$$\lambda_{\max} \chi[A_1, \dots, A_k] < \frac{3\sqrt{2}}{\sqrt{k}}. \quad (4.2)$$

For any zero-diagonal Hermitian matrix L , without a norm restriction,

$$\lambda_{\max} \chi[L, -L] \geq \frac{\|L\|}{2}. \quad (4.3)$$

Consequently, on a nonempty finite set, if every site is unassigned, the unit-weight polynomial without diagonal shifts has largest root $< 3/\sqrt{r}$. If every site is assigned, that unit-weight unshifted polynomial is

$$\prod_{i=1}^r \chi[T|_{c^{-1}(i)}, -T|_{c^{-1}(i)}],$$

where factors on empty color classes are 1.

Proof. The normalization agrees with Definition 5 of [RS17]. Real-rootedness is part of the mixed determinantal theory and also follows from Lemma 4.2 below. Equation (4.2) is [RS17, Theorem 25], whose hypotheses are zero-diagonal Hermitian contractions and $k \geq 2$.

By [RS17, Proposition 16 and Theorem 15], for a tuple of zero-diagonal Hermitian matrices, replacing one member by zero cannot increase its largest root. Expansion in principal minors gives

$$\chi[L, 0](z) = \det(zI - L/2) :$$

a fixed principal minor on S occurs with probability $2^{-|S|}$. Applying monotonicity while retaining either L or $-L$ proves Equation (4.3). The all-unassigned case uses the tuple containing r copies each of T and $-T$, so $k = 2r$. For a fully assigned coloring, independence between the color classes gives the stated factorization. $\square$

4.2 Stable quotients and rooted spectral measures

The next lemma provides the interlacing needed to measure the effect of one site. Its half-plane assertion also allows non-Hermitian matrices: this extension will later control analytic perturbations of the data. For a matrix B , write $\text{Im } B = (B - B^*)/(2i)$.

Lemma 4.2 (Polynomial quotients). *For the finite data above,*

$$P_I(Z) = \left(\prod_{v \in I} \frac{1}{k_v!} \partial_{z_v}^{k_v-1} \right) \prod_j \det((Z - B_j)|_{U_j}), \quad \partial_{z_v} P_I = P_{-v}. \quad (4.4)$$

More generally, use any matrices B_j on U_j in the same independent-choice construction. If $\text{Im } B_j \leq hI$ for a real number h , then $P_I(Z)$ and every deletion polynomial are nonzero whenever $\text{Im } z_v > h$ at every remaining site. In this region,

$$\text{Im } \frac{P_I(Z)}{P_{-v}(Z|_{I \setminus \{v\}})} \geq \text{Im } z_v - h. \quad (4.5)$$

Return to the Hermitian data in Equation (4.1). For $v \in I$, set

$$Q_v = P_{-v}, \quad F_v(z) = \frac{P_I(z)}{Q_v(z)}, \quad G_v(z) = \frac{Q_v(z)}{P_I(z)}.$$

These rational functions are analytic in the upper half-plane, and

$$F_v(z) = z + a_v - \sum_{\lambda} \frac{b_{v,\lambda}}{z - \lambda}, \quad b_{v,\lambda} \geq 0, \quad (4.6)$$

where λ ranges over the distinct roots of Q_v . Let Λ_v be the diagonal matrix listing those roots with multiplicity. There is a vector g_v , with squared coordinate $b_{v,\lambda}$ on one copy of each λ and zero on its other copies, such that

$$H_v = \begin{pmatrix} -a_v & g_v^* \\ g_v & \Lambda_v \end{pmatrix} \quad (4.7)$$

has characteristic polynomial P_I . Its first-coordinate spectral measure ρ_v is a probability measure satisfying

$$G_v(z) = \int_{\mathbb{R}} \frac{1}{z - x} \rho_v(dx), \quad \sum_{v \in I} \rho_v = \sum_{P_I(x)=0} \delta_x. \quad (4.8)$$

The roots on the right are counted with multiplicity, and Q_v interlaces P_I .

Proof. Each determinant factor is affine in any of its diagonal variables. When differentiating $k_v - 1$ times in z_v , every surviving Leibniz term leaves v in exactly one of its k_v available factors and deletes it from the others. The multiplicity $(k_v - 1)!$, divided by $k_v!$, gives probability $1/k_v$. Processing all vertices gives the outcome average, and differentiation once more in z_v deletes that site. This proves Equation (4.4), including for the more general matrices.

If $\text{Im } z_v > h$, each matrix $Z - B_j$ on U_j has strictly positive imaginary part and is invertible. Its determinant is therefore nonzero. Differentiation preserves nonvanishing in the specified half-plane by the Gauss–Lucas Theorem. Here the degree cannot drop at a specialization of the other variables: after any subset of the vertex operations has been performed, the leading coefficient of degree k_v in an unprocessed variable z_v is the corresponding partially processed expression with v deleted. Induct simultaneously over the processed vertices and all principal restrictions. That leading coefficient is nonzero by the induction hypothesis, so every required derivative has its indicated degree and is nonzero in the half-plane. This also proves the assertion for all deletion polynomials.

For fixed other coordinates in that region, $P_I/P_{-v} = z_v + \zeta_0$, by the derivative identity and multiaffineness. Its zero has imaginary part at most h . This gives Equation (4.5).

For Hermitian data, the uniform-variable polynomials have real coefficients; half-plane nonvanishing and conjugation therefore show that all their roots are real. Equation (4.5) with $h = 0$ implies that $F_v(z) - z$ has nonnegative imaginary part. Its constant at infinity is a_v , since every outcome has v diagonal $-a_v$. A nonremovable real pole of a real rational function with nonnegative imaginary part must be simple with nonpositive residue: approach the pole from different angles in the upper half-plane to exclude higher orders, and then approach vertically to determine the sign. This proves Equation (4.6).

The Schur complement of $z - \Lambda_v$ in $z - H_v$ is $F_v(z)$. Thus $\det(z - H_v) = Q_v(z)F_v(z) = P_I(z)$, and its first resolvent entry is $1/F_v(z) = G_v(z)$. The spectral theorem proves the first identity in Equation (4.8), and principal-matrix interlacing proves the interlacing assertion. Finally, uniform specialization of Equation (4.4) gives

$$\sum_{v \in I} G_v(z) = \frac{P'_I(z)}{P_I(z)}.$$

The right side is the Cauchy transform of the root-counting measure. Uniqueness of partial fractions proves the second identity in Equation (4.8). $\square$

We will compare two modifications at an unassigned site v . For $i \in \{1, \dots, r\}$, first condition its pair on color i , keeping both signs uniformly distributed and retaining its shift $\tau_v = t$. Denote the resulting polynomial and quotient by $P_{v,i}^0$ and $F_{v,i}^0 = P_{v,i}^0/Q_v$. Next increase that site's shift to βt , as required by a permanent assignment. Denote these by $P_{v,i}$ and $F_{v,i}$. All other site data remain fixed, and deletion of v gives the same polynomial Q_v in each case. Consequently

$$\frac{1}{r} \sum_{i=1}^r F_{v,i}^0 = F_v, \quad F_{v,i} = F_{v,i}^0 + (\beta - 1)tW_v. \quad (4.9)$$

The arrow representation and rooted measures apply to these modified data as well. In particular the residues of $F_{v,i}^0$ are nonnegative and average to the residues of F_v .

4.3 Positive-root costs and locality

For a monic real-rooted polynomial P , define

$$s(P) = \sum_{P(x)=0} x_+, \quad x_+ = \max\{x, 0\}, \quad s(1) = 0.$$

The roots in this sum are counted with multiplicity. For the finite site data, define the rooted cost and the deletion increment by

$$m_v = \int x_+ \rho_v(dx), \quad d_v = s(P_I) - s(P_{-v}). \quad (4.10)$$

Thus $\sum_v m_v = s(P_I)$. Superscripts $0, i$ and i will indicate, respectively, the conditioned and permanently assigned options at v . For example, $d_v^i = s(P_{v,i}) - s(Q_v)$.

The initial shift $t > 3/\sqrt{r}$ gives zero cost before any assignment. For a fully assigned block, excessive pinched norm forces positive cost. The following finite statement makes these two roles precise.

Lemma 4.3 (Initial cost and detection of excessive norm). *For the finite data of this section, the following hold.*

1. *If every site is unassigned and $t > 3/\sqrt{r}$, then $P_I(x) > 0$ for every $x \geq 0$, and $s(P_I) = 0$. This includes $I = \emptyset$, with $P_\emptyset = 1$.*
2. *Suppose $I \neq \emptyset$ and every site is assigned, and put*

$$T_c = \sum_{i=1}^r 1_{\{c=i\}} T 1_{\{c=i\}}.$$

For every $\alpha > 0$, if $\|T_c\| > 2\beta t + \alpha$, then

$$\lambda_{\max} P_I \geq \frac{\alpha}{2} \min_{v \in I} W_v, \quad s(P_I) \geq \frac{\alpha}{2} \min_{v \in I} W_v.$$

Proof. For $J \subseteq I$, let $P_J^{\text{un}}(Z)$ be the expected polynomial with the same pair laws, all weights equal to one, and all diagonal shifts removed. Factoring the weight matrix from every outcome determinant gives

$$P_J(x) = \left(\prod_{v \in J} W_v \right) P_J^{\text{un}}((\tau_v + x/W_v)_{v \in J}). \quad (4.11)$$

Both sides equal one for $J = \emptyset$. We also use the following consequence of multiaffineness and the deletion identity in Equation (4.4). For real coordinates $b = (b_v)_{v \in I}$ and increments $h_v \geq 0$,

$$P_J^{\text{un}}((b+h)|_J) = \sum_{S \subseteq J} P_{J \setminus S}^{\text{un}}(b|_{J \setminus S}) \prod_{v \in S} h_v.$$

Thus positivity of all principal evaluations at b is preserved when any coordinates are increased.

If every site is unassigned and $t > 3/\sqrt{r}$, Lemma 4.1 puts t strictly above the largest root of every nonempty $P_J^{\text{un}}(z)$. All principal evaluations at the common coordinate t are therefore positive. For $x \geq 0$, the coordinates $t + x/W_v$ in Equation (4.11) are at least t , so $P_I(x) > 0$. Real-rootedness then gives $s(P_I) = 0$.

Now suppose every site is assigned and $\|T_c\| > 2\beta t + \alpha$. The fully assigned factorization and the lower bound in Lemma 4.1 show that $P_I^{\text{un}}(z)$ has a root greater than $\beta t + \alpha/2$. Set

$x_0 = (\alpha/2) \min_{v \in I} W_v > 0$. If the largest root of P_I were smaller than x_0 , repeated interlacing would put x_0 strictly above every root of every nonempty principal restriction. Thus $P_J(x_0) > 0$ for all $J \subseteq I$. By Equation (4.11), all the corresponding unweighted principal evaluations are positive at

$$b_v = \beta t + x_0/W_v \leq \beta t + \alpha/2.$$

Increasing these coordinates to any common value $q \geq \beta t + \alpha/2$ preserves that positivity. In particular $P_I^{\text{un}}(q) > 0$ for every such q , contradicting the root above $\beta t + \alpha/2$. Hence the largest root of P_I is at least x_0 , which also gives the stated lower bound for $s(P_I)$. $\square$

Lemma 4.4 (Deletion costs and their tails). *For the finite Hermitian data above, let $n_P(x) = \#\{\lambda : P(\lambda) = 0, \lambda > x\}$, with multiplicity. Then, for every $v \in I$,*

$$0 \leq n_{P_I}(x) - n_{Q_v}(x) \leq 1, \quad (4.12)$$

$$d_v = \int_0^\infty (n_{P_I} - n_{Q_v})(x) dx, \quad (4.13)$$

$$\int_B^\infty (n_{P_I} - n_{Q_v})(x) dx \leq 2 \int (x - B)_+ \rho_v(dx) + (a_v + B) \rho_v((B, \infty)), \quad B \geq 0. \quad (4.14)$$

In particular $0 \leq d_v \leq 2m_v + a_v$. All these conclusions hold for either modification at v , with its own shift and rooted measure.

Proof. Equation (4.12) follows from the principal interlacing in Lemma 4.2; integration gives Equation (4.13). To prove the tail estimate, write $H = H_v$, $\Lambda = \Lambda_v$, $a = a_v$, and $L = H - BI$. Let E be the first-coordinate projection, $D = 1 - E$, and $R = 1_{(0, \infty)}(L)$. The compression DRD is a positive contraction on the deletion coordinates. The variational formula for the positive trace therefore gives

$$\text{Tr}(\Lambda - BI)_+ \geq \text{Tr}((\Lambda - BI)DRD).$$

Since $L - DLD = EL + LE - ELE$, it follows that

$$\begin{aligned} \text{Tr}(H - BI)_+ - \text{Tr}(\Lambda - BI)_+ &\leq \text{Tr}((EL + LE - ELE)R) \\ &= 2\langle e_0, LRe_0 \rangle + (a + B)\langle e_0, Re_0 \rangle \\ &= 2 \int (x - B)_+ \rho_v(dx) + (a + B) \rho_v((B, \infty)). \end{aligned}$$

Here e_0 is the first unit vector of the arrow matrix. The left side is the tail integral in Equation (4.14). Taking $B = 0$ and using that ρ_v is a probability measure gives the last assertion. $\square$

The deletion increment is defined using all roots of a polynomial, whereas m_v is attached to one site. We next record the locality which relates these finite quantities to graph neighborhoods. A graph supporting T means an undirected graph on I , without loops, such that $T_{uv} = 0$ whenever $u \neq v$ are not adjacent.

Lemma 4.5 (Rooted locality and bounds). *Let a graph support the finite zero-diagonal Hermitian contraction T . The coefficient of z^{-1-k} in $G_v(z)$ is $\int x^k \rho_v(dx)$ and is determined by the data on the radius- k ball about v . The Laurent coefficients of $F_v(z)$, and of the two options $F_{v,i}^0(z)$ and $F_{v,i}(z)$ at an unassigned site, likewise depend only on a radius bounded by their order. Moreover,*

$$\int x^2 \rho_v(dx) = a_v^2 + \sum_{u \neq v} W_v W_u |T_{vu}|^2 \mathbb{P}(\omega_u = \omega_v), \quad \sum_{\lambda} b_{v,\lambda} = \sum_{u \neq v} W_v W_u |T_{vu}|^2 \mathbb{P}(\omega_u = \omega_v). \quad (4.15)$$

The same identities hold for either site option, using the corresponding law of the pairs and diagonal shift.

If neighboring weights have ratios in $[K^{-1}, K]$, $K \geq 1$, then for all these data

$$\int x^2 \rho_v(dx) \leq ((\beta t)^2 + K)W_v^2, \quad m_v \leq C_0 W_v, \quad d_v \leq (2C_0 + \beta t)W_v, \quad C_0 = \sqrt{(\beta t)^2 + K}. \quad (4.16)$$

If all weights are at most H_0 , every polynomial and every principal restriction has all its roots in

$$[-L, L], \quad L = (1 + \beta t)H_0. \quad (4.17)$$

The rooted measures have the same support bound.

Proof. Put $u_v = z_v^{-1}$ and consider the formal polynomial

$$\mathcal{H}_I(u) = P_I(Z) \prod_{v \in I} u_v = \sum_{S \subset I} c_S \prod_{v \in S} u_v.$$

The coefficient c_S is the expected determinant of the negative outcome matrix restricted to S . If S splits into graph components, the restricted outcome matrix splits into the same blocks, and their choices are independent. Hence c_S factors over the components. Since $\mathcal{H}_I(0) = 1$, its formal logarithm is defined. A coefficient in this logarithm depends only on its supported subset. On a disconnected subset $\mathcal{H}_I$ factors, so its logarithm is a sum over components. Thus every nonconstant monomial in $\log \mathcal{H}_I$ has connected support.

By Equation (4.4),

$$G_v(Z) = \partial_{z_v} \log P_I(Z).$$

Apart from the term $1/z_v$, its monomials arise from connected supports containing v . After uniform specialization, the coefficient of order z^{-1-k} therefore uses only the radius- k ball at v . The Cauchy-transform expansion identifies that coefficient with the k -th rooted moment. Formal inversion of the series with leading term $1/z$ gives the same locality statement for F_v . Conditioning or changing the shift at v preserves this argument.

For singleton and two-point sets the coefficients are

$$c_{\{v\}} = a_v, \quad c_{\{u,v\}} = a_u a_v - W_u W_v |T_{uv}|^2 \mathbb{P}(\omega_u = \omega_v).$$

Expanding the logarithmic derivative to order z^{-3} gives the first identity in Equation (4.15). Alternatively, the arrow matrix has first diagonal $-a_v$ and off-diagonal squared norm $\sum_\lambda b_{v,\lambda}$; comparison with its second rooted moment gives the second identity.

Under the neighboring-ratio hypothesis, the sum in Equation (4.15) is at most

$$KW_v^2 \sum_{u \neq v} |T_{vu}|^2 \leq KW_v^2.$$

This holds for each conditioned law as well, since every collision probability is at most one. Also $a_v \leq \beta t W_v$. Cauchy–Schwarz for the probability measure ρ_v , followed by Lemma 4.4, proves Equation (4.16).

Finally every outcome has norm at most $(1 + \beta t)H_0$, because principal compression and pinching preserve the contraction bound of T . For real $z > L$, all its characteristic determinants are positive; for $z < -L$, all have sign $(-1)^{|I|}$. Their average therefore has no root outside $[-L, L]$. Real-rootedness proves Equation (4.17). The same reasoning applies after restriction or a site option, and the spectral representation gives the assertion for rooted measures. $\square$

4.4 The single-site assignment estimate

We now compare the cost of leaving a site unassigned with the costs of permanently assigning its possible colors. The comparison depends only on the arrow matrix and the averaging of its nonnegative residues. This is the finite estimate that will govern the coloring procedure.

Lemma 4.6 (Assigning one color). *For a finite zero-diagonal Hermitian contraction T , arbitrary positive weights, $r \geq 1$, $t > 0$, and a partial coloring as above, let v be unassigned. With $\beta = 80$, there is $i \in \{1, \dots, r\}$ such that*

$$d_v^i - d_v \leq 24m_v. \quad (4.18)$$

The bound is independent of the finite set and its weights.

Proof. Write $a = a_v = tW_v > 0$, $m = m_v$, $d = d_v$, and

$$H = \begin{pmatrix} -a & g^* \\ g & \Lambda \end{pmatrix}$$

for the old arrow matrix. By Equation (4.9), the color-conditioned residues $b_{i,\lambda}$ are nonnegative and have average b_λ . A permanent assignment replaces these residues by $b_{i,\lambda}$ and the first diagonal by $-\beta a$. All deletion eigenvalues Λ stay fixed.

Separate the deletion coordinates into those with $\lambda > 0$ and those with $\lambda \leq 0$, and use subscripts $+$ and $-$ for these two blocks. Write $H = H_+ - H_-$ for its positive and negative parts; subscripts on $H_\pm$ below specify compressions to the deletion blocks, not additional positive or negative parts. Set

$$e_+ = \text{Tr}(H_-)_{++}, \quad e_- = \text{Tr}(H_+)_{--}.$$

Since the first diagonal of H_+ is m , comparison of the diagonal blocks in $H = H_+ - H_-$ gives

$$d = m + e_+ + e_-, \quad (H_-)_{00} = m + a. \quad (4.19)$$

We treat separately the cases in which the rooted cost is at least the shift and in which it is smaller.

Case $m \geq a$. Consider the two side arrows obtained by retaining the first coordinate and just one deletion block, initially with first diagonal $-a$. For the positive block there is exactly one negative eigenvalue, say $-L_0$, and

$$L_0 = a + \sum_{\lambda > 0} \frac{b_\lambda}{L_0 + \lambda}, \quad d^+ = L_0 - a.$$

Here d^+ is its positive-trace increment over its deletion block. The formulas remain valid for an empty positive block, with $L_0 = a$. For the nonpositive block there is at most one positive eigenvalue. Let s_0 denote that eigenvalue if present, and put $s_0 = 0$ otherwise. Its increment is $d^- = s_0$, and

$$\sum_{\lambda \leq 0} \frac{b_\lambda}{s_0 + |\lambda|} \leq s_0 + a. \quad (4.20)$$

If $s_0 = 0$, all residues at $\lambda = 0$ must vanish; otherwise the side quotient would have a positive zero. Omit those terms in Equation (4.20). If the remaining sum at zero were greater than a , continuity and the behavior at infinity would again produce a positive zero, which proves the displayed inequality also in this case.

Pinching off the other coupling in H cannot increase its positive trace. The uncoupled block contributes exactly its own positive trace, so $d^+ \leq d$ and $d^- \leq d$. By averaging the residues, at least three quarters of the colors satisfy

$$\sum_{\lambda > 0} \frac{b_{i,\lambda}}{L_0 + \lambda} \leq 4d^+,$$

and at least three quarters satisfy

$$\sum_{\lambda \leq 0} \frac{b_{i,\lambda}}{s_0 + |\lambda|} \leq 4(s_0 + a).$$

Thus some color satisfies both bounds. A zero old residue is zero for every color by nonnegativity, so the omitted zero-denominator terms remain absent.

For this color, the positive-block arrow with first diagonal $-a$ has increment at most $4d^+$. Indeed, if its negative eigenvalue magnitude is at most L_0 , the assertion is immediate; otherwise its root equation bounds the increment by the preceding test sum. The nonpositive-block arrow with first diagonal $-4a$ has increment at most $4s_0$. If it had a positive root $s > 4s_0$, its decreasing residue sum would be at most its value at s_0 , hence at most $4(s_0 + a)$, whereas its root equation would require that sum to be $s + 4a > 4(s_0 + a)$.

Extend the two side arrows by zero and add them. The resulting full arrow has first diagonal $-5a$. Subadditivity of positive trace, which follows from its variational formula, shows that its increment is at most $4d^+ + 4d^-$. Decreasing the first diagonal further to $-\beta a$ can only decrease positive trace. Therefore

$$d_v^i \leq 4(d^+ + d^-) \leq 8d, \quad d_v^i - d \leq 7d \leq 7(2m + a) \leq 21m,$$

where Lemma 4.4 was used in the last step.

Case $m < a$. The positive and negative parts of H give more useful residue bounds in this case. For a positive block matrix, the outer product of its first coupling column is bounded by its first diagonal entry times its remaining block. Apply this to H_+ and H_- , and use $(p - q)(p - q)^* \leq 2pp^* + 2qq^*$ for the difference of their coupling columns. As $(H_+)_{++} = \Lambda_+ + (H_-)_{++}$ and $(H_-)_{--} = |\Lambda_-| + (H_+)_{--}$, this gives

$$g_+ g_+^* \leq 2m\Lambda_+ + (4m + 2a)(H_-)_{++}, \quad (4.21)$$

$$g_- g_-^* \leq 2(m + a)|\Lambda_-| + (4m + 2a)(H_+)_{--}. \quad (4.22)$$

Conjugate Equation (4.21) by $(32a + \Lambda_+)^{-1/2}$ and take the operator norm. The left side has rank at most one, so its norm is the corresponding residue sum. The first term on the right has norm at most $2m$, and the error term has norm at most its trace. Since $4m + 2a \leq 6a$,

$$\sum_{\lambda > 0} \frac{b_\lambda}{32a + \lambda} \leq 2m + \frac{6}{32}e_+. \quad (4.23)$$

Put $L_1 = e_-$. If $L_1 > 0$, conjugating Equation (4.22) by $(L_1 + |\Lambda_-|)^{-1/2}$ in the same way gives

$$\sum_{\lambda \leq 0} \frac{b_\lambda}{L_1 + |\lambda|} \leq 2(m + a) + (4m + 2a) \leq 10a. \quad (4.24)$$

Indeed the trace of $(H_+)_{--}$ is L_1 , so its conjugated norm is at most 1. If $L_1 = 0$, positivity implies $(H_+)_{--} = 0$. Equation (4.22) then forces zero coupling on the kernel of $|\Lambda_-|$. Use its inverse only on its nonzero support and omit the zero residues at $\lambda = 0$; Equation (4.24) remains valid.

Choose a color for which both residue sums in Equations (4.23) and (4.24) are at most four times their indicated upper bounds. Such a color exists by the same averaging argument. For that color, the positive-block arrow with diagonal $-32a$ has negative eigenvalue magnitude at least $32a$; its root equation therefore bounds its positive-trace increment by

$$8m + \frac{24}{32}e_+.$$

The nonpositive-block arrow with diagonal $-40a$ has increment at most L_1 . Indeed a positive root $s > L_1$ would have residue sum at most its test value $40a$, whereas its root equation requires $s + 40a > 40a$. This also covers $L_1 = 0$, with zero residues omitted as above.

Adding the side arrows gives the full arrow with diagonal $-72a$. Subadditivity of positive trace and then decreasing the diagonal to $-80a$ yield

$$d_v^i \leq 8m + \frac{3}{4}e_+ + e_-.$$

Together with Equation (4.19), this gives

$$d_v^i - d \leq 7m - \frac{1}{4}e_+ \leq 7m.$$

Both cases imply Equation (4.18). All arguments allow empty deletion blocks and empty sums, so they include a one-site matrix and every zero-residue case. $\square$

5 Rooted costs on invariant measured graphs

The preceding finite polynomial estimates control the effect of assigning a single vertex. We now pass from finite sets to measured graphs, where many vertices must be changed simultaneously. There are two points to establish. First, the integral of the rooted positive cost changes by the integral of the single-vertex deletion increments, evaluated in an ordered sequence of comparison environments. Second, deleting vertices outside a fixed weight band has a small effect on the transforms within that band, with a bound independent of the largest weight anywhere in the graph. The latter uniformity will allow us to remove weight cutoffs in the next section.

Let (X, ν) be a standard finite measure space, and let $\mathcal{R}$ be a countable measured equivalence relation preserving ν . Thus each measurable partial isomorphism whose graph lies in $\mathcal{R}$ preserves measure. Let $G \subset \mathcal{R}$ be a measurable undirected graph without loops, of degree at most $D_0 < \infty$. On its orbit spaces suppose that we are given measurable Hermitian matrices T of norm at most one, with zero diagonal and entries supported on G . As before, changing the root may conjugate these matrices by diagonal unitaries. All the constructions below are invariant under those conjugacies. Null exceptions are discarded along their $\mathcal{R}$ -saturation.

Fix an integer $r \geq 1$, $t > 0$, and $\beta = 80$. A partial coloring is a measurable map $c : X \rightarrow \{0, 1, \dots, r\}$, where 0 denotes an unassigned vertex. Use the shifts and polynomial choices of Section 4, and let $W : X \rightarrow (0, H_0]$ be a measurable weight, where $H_0 < \infty$. When a measurable set of vertices has been deleted, take the principal restrictions of the graph and matrix, retaining the original graph to measure distances. At absent vertices every rooted cost or transform is set equal to zero, and the rooted measure is the zero measure. On a finite principal set I we use the notation

$$P_I, \quad Q_{I,v} = P_{I \setminus \{v\}}, \quad F_{I,v} = \frac{P_I}{Q_{I,v}}, \quad G_{I,v} = \frac{1}{F_{I,v}}, \quad \rho_{I,v}, \quad m_{I,v}, \quad d_{I,v}$$

from the preceding section. In particular, with roots counted with multiplicity,

$$m_{I,v} = \int x_+ \rho_{I,v}(dx), \quad d_{I,v} = s(P_I) - s(Q_{I,v}), \quad s(P) = \sum_{P(x)=0} x_+.$$

We write $a_v = \tau_v W_v$ and set

$$L = (1 + \beta t)H_0. \tag{5.1}$$

All roots and rooted spectral measures in these finite calculations lie in $[-L, L]$, including the virtual color choices at an unassigned vertex.

Measure invariance gives the mass transport identity

$$\int_X \sum_{u \in [v]_{\mathcal{R}}} h(v, u) d\nu(v) = \int_X \sum_{v \in [u]_{\mathcal{R}}} h(v, u) d\nu(u) \quad (5.2)$$

for nonnegative measurable h , and for integrable complex-valued h . Indeed, partition $\mathcal{R}$ into countably many graphs of partial one-to-one maps and change variables on each graph.

5.1 Limits on bounded weighted graphs

Lemma 5.1. *For the invariant measured graph data on the finite measure space (X, ν) specified above, the rooted measures, transforms, and costs have limits as finite principal subsets exhaust the graph component of the root. Denote them by ρ_v , G_v , F_v , m_v , and d_v . They are independent of the exhaustion and are measurable in v . At each present vertex, ρ_v is a probability measure supported on $[-L, L]$, and, for z in the open upper half-plane,*

$$G_v(z) = \int \frac{1}{z - x} \rho_v(dx), \quad F_v(z) = \frac{1}{G_v(z)}.$$

The same assertions hold after conditioning the color at one unassigned vertex, with either its original shift or its increased shift.

For fixed D_0, H_0, r, t , the costs m_v, d_v and the virtual shifted costs d_v^i , $1 \leq i \leq r$, are uniformly approximable by calculations on a ball of finite radius about v . The approximation radius is independent of the particular graph, weights, partial coloring, and set of deleted vertices. These statements apply to a finite-measure restriction of an invariant measured relation as well.

Proof. By Lemma 4.5, each moment $\int x^k \rho_{I,v}(dx)$ is determined by a bounded neighborhood of v , and is unchanged once I contains that neighborhood. Compactness of the probability measures on $[-L, L]$ and uniqueness from their moments therefore give an exhaustion-independent weak limit ρ_v . Its Cauchy transform is the limit of $G_{I,v}$, locally uniformly on the upper half-plane. Moreover,

$$-\operatorname{Im} G_v(z) = (\operatorname{Im} z) \int |z - x|^{-2} \rho_v(dx) > 0,$$

so this limit can be inverted. Hence $F_{I,v}$ converges to F_v there. The same argument applies to the altered color law or shift at the root. The finite quotients also have the form

$$F_{I,v}(z) - z - a_v = - \int \frac{1}{z - x} \kappa_{I,v}(dx),$$

where the positive residue measure $\kappa_{I,v}$ is supported on $[-L, L]$. Its mass is at most H_0^2 by the second-moment formula in Lemma 4.5; the same bound holds for the conditional color laws. Compactness and uniqueness of Cauchy transforms therefore give a limiting residue measure as well. This also justifies taking the local Laurent-coefficient limits of the quotients below. Since x_+ is continuous on $[-L, L]$, the costs $m_{I,v}$ converge as well. Polynomial approximation on this interval, together with moment locality, makes this convergence uniformly approximable by finite neighborhood calculations.

Deletion increments require a different approximation, because their definition involves the roots at every vertex of a finite set. For a real polynomial f , put

$$d_{I,v}[f] = \sum_{P_I(x)=0} f(x) - \sum_{Q_{I,v}(x)=0} f(x).$$

The identity that the sum of the rooted measures counts all polynomial roots gives

$$d_{I,v}[f] = \sum_{u \in I} \left(\int f d\rho_{I,u} - 1_{\{u \neq v\}} \int f d\rho_{I \setminus \{v\},u} \right). \quad (5.3)$$

Each summand is local. It vanishes unless u lies within a radius depending only on $\deg f$ of v . Thus the entire increment can be computed in a sufficiently large finite neighborhood of v , independently of the ambient finite set. This defines its value on the infinite graph.

To pass from polynomials to x_+ , for an absolutely continuous real function f on $[-L, L]$ use interlacing in the form

$$d_{I,v}[f] = f(-L) + \int_{-L}^L f'(x)(n_{P_I}(x) - n_{Q_{I,v}}(x)) dx, \quad 0 \leq n_{P_I}(x) - n_{Q_{I,v}}(x) \leq 1. \quad (5.4)$$

Here $n_P(x)$ counts roots strictly greater than x . The formula follows by writing $f(\lambda) = f(-L) + \int_{-L}^\lambda f'(x) dx$ for each root; the difference in degrees is one. Consequently

$$|d_{I,v}[f] - d_{I,v}[g]| \leq |f(-L) - g(-L)| + \|f' - g'\|_{L^1([-L, L])}. \quad (5.5)$$

Approximate f' in L^1 by polynomials and integrate them with prescribed initial value $f(-L)$. This gives polynomial approximants with both uniform function error and the increment error in Equation (5.5) tending to zero. Apply this to $f(x) = x_+$. The local polynomial increments are eventually constant, and their approximation error is uniform in every finite set. Hence $d_{I,v}$ has the stated limit and uniform locality property. The same proof works for each of the finitely many virtual options at v .

Finally, finite balls in a measurable bounded-degree graph, their finite matrix data, and their conjugacy-invariant polynomial calculations are measurable. The limiting constructions above are therefore measurable. Deleting vertices or restricting the relation does not change the argument: use the original graph for the locality radius and the retained vertices for each principal calculation. $\square$

5.2 Integral change under ordered updates

We next compare two global partial colorings, allowing deletion of vertices. The order in which vertices are compared need not have a first element. Only finitely many comparisons occur in any given polynomial local calculation, which is enough for the following identity.

Lemma 5.2. *Under the bounded finite-measure hypotheses of Lemma 5.1, let the old and new data differ only by changes of partial coloring or deletion at a measurable set E_0 of vertices. Fix a measurable real-valued function that is injective on E_0 in each relation class, and order these vertices by its values. At $u \in E_0$, use new data at all earlier vertices and old data at all later vertices. Let δd_u be the new-minus-old difference of the single-vertex deletion cost at u in this environment, with cost zero when u is absent. Then*

$$\int_X (m_v^{\text{new}} - m_v^{\text{old}}) d\nu(v) = \int_{E_0} \delta d_u d\nu(u). \quad (5.6)$$

In particular, deletion alone cannot increase the integral of m_v .

Proof. An injective measurable point code supplies such an ordering if none has already been specified. Begin with a real polynomial f in place of x_+ . Write $m_v[f] = \int f d\rho_v$ at present vertices and zero at absent ones. For each root v , this quantity depends only on a fixed finite-radius neighborhood. List the finitely many vertices of E_0 in that neighborhood in their specified order and

telescope its change over this list. Each term agrees with the change produced at that vertex in the global comparison environment: changes outside the neighborhood have no effect on this polynomial rooted datum. Thus no well-ordering or infinite telescoping at an individual root is being assumed.

Integrate the resulting finite local sum and apply Equation (5.2) to interchange the root and the changed vertex. For a fixed changed vertex u , the sum over affected roots equals the change in $d_u[f]$. Indeed, this can be checked on a sufficiently large finite principal neighborhood using Equation (5.3): the old and new polynomials have the same deletion polynomial at u . If the vertex is deleted, the new rooted datum there and the new deletion increment are both zero by convention. Bounded degree, bounded spectral support, and finite total measure justify all integrals for this fixed polynomial. We have proved

$$\int_X (m_v^{\text{new}}[f] - m_v^{\text{old}}[f]) d\nu(v) = \int_{E_0} \delta d_u[f] d\nu(u).$$

Choose the polynomial approximants to x_+ constructed in the proof of Lemma 5.1. Uniform function approximation controls the two rooted terms on the left; the L^1 derivative bound in Equation (5.5) controls both comparison increments on the right. Finite measure then permits passage to the limit and proves Equation (5.6). For a deletion the comparison increment is $-d_u \leq 0$, by interlacing. This proves the last assertion. $\square$

5.3 Deletion estimates independent of the largest weight

The preceding arguments allow constants depending on H_0 . To pass later to unbounded weights, we need a comparison on a fixed band that is uniform as H_0 increases. Cauchy transforms provide this comparison. Besides G_v , we must control the conditional color choices used in the single-vertex assignment inequality.

At an unassigned vertex v , let $F_{v,i}^0$ be the quotient obtained by conditioning its color to be i while leaving its shift equal to t . The permanent option has quotient

$$F_{v,i}(z) = F_{v,i}^0(z) + (\beta - 1)tW_v.$$

Define the auxiliary analytic field

$$J_i(v, z) = (F_{v,i}^0(z) - z - a_v)G_v(z), \quad c(v) = 0 \text{ and } v \text{ present.} \quad (5.7)$$

Set $J_i(v, z) = 0$ at all other vertices. At present unassigned vertices the reciprocal transforms here are nonzero by Lemma 5.1. The residue formula of Lemma 4.5 shows that these analytic fields are bounded on upper-half-plane compact sets when the root weight and its neighbors' weight ratios are bounded.

Lemma 5.3. *Consider the invariant measured graph data on the finite measure space (X, ν) specified above and assume, in addition, that neighboring weights have ratios in $[K^{-1}, K]$, where $K \geq 1$. Fix $0 < b_1 < b_2 < \infty$ and the band*

$$\mathcal{U} = \{v : b_1 \leq W_v \leq b_2\}.$$

Delete a measurable set E_{hi} disjoint from $\mathcal{U}$, without changing the coloring at retained vertices. Let δ_{tot} denote the difference between the original and the deleted configurations, in either orientation. For each compact set Ω in the open upper half-plane there is a constant C such that, for every $z \in \Omega$ and $1 \leq i \leq r$,

$$\int_{\mathcal{U}} |\delta_{\text{tot}} G_v(z)| d\nu(v) + \int_{\mathcal{U} \cap \{c=0\}} |\delta_{\text{tot}} J_i(v, z)| d\nu(v) \leq C\nu(E_{\text{hi}}). \quad (5.8)$$

Only band vertices present before deletion are included in these integrals. The constant may depend on $\Omega, r, t, \beta, D_0, K, b_1, b_2$, but not on H_0 , the measure or its total mass, the partial coloring, the graph itself, or the deleted set.

Proof. We may discard initially absent vertices from E_{hi} : this leaves both configurations unchanged and only decreases its measure. Thus every deletion vertex below is present before the deletions begin. We first prove bounds for one deletion in a finite principal set. We then pass these bounds to infinite bounded graphs and integrate them using an analytic counterpart of the ordered-change identity.

A single deletion and the field G . Fix a finite principal set I , $u \in I$, and z with $y = \text{Im } z > 0$. For arbitrary complex numbers λ_v satisfying $|\lambda_v| \leq 1$, perturb the diagonal variables to $z + \zeta \lambda_v$, $v \in I$. On the disk $|\zeta| < y/2$ they remain in the upper half-plane. By Lemma 4.2, the quotient $P_I/P_{I \setminus \{u\}}$ in these variables has positive imaginary part. Its logarithm has a holomorphic branch with imaginary part in $(0, \pi)$ throughout the disk.

We use the elementary derivative estimate that a holomorphic function on a disk of radius R , with bounded imaginary part, has derivative at the center bounded by a numerical constant times the oscillation of that imaginary part divided by R . To see this, restrict to a concentric circle of radius $R/2$ and recover the first Taylor coefficient from the first Fourier coefficient of the imaginary part. Applied to the logarithm just described, it bounds its derivative at zero by C_0/y . Logarithmic differentiation of P_I in a diagonal variable gives $G_{I,v}$, so this derivative is

$$\sum_{v \in I} \lambda_v (G_{I,v}(z) - G_{I \setminus \{u\},v}(z)),$$

where the second transform at $v = u$ is zero. Choosing the phases of λ_v to make all summands nonnegative real yields

$$\sum_{v \in I} |G_{I,v}(z) - G_{I \setminus \{u\},v}(z)| \leq C_0/y. \quad (5.9)$$

This estimate contains no bound on the weights.

A single deletion and the conditional color field. Now assume $u \notin \mathcal{U}$, and take $\lambda_v = 0$ except at unassigned vertices in $\mathcal{U}$. Perturb the off-diagonal parts of both pair matrices of color i by left and right diagonal scaling with entries $1 + \zeta \lambda_v$. There is no conjugation of ζ in the right scaling, and the fixed diagonal shift is left unchanged. This gives an analytic perturbation, though its matrices need not be Hermitian for complex ζ .

Let $D_\lambda = \text{diag}(\lambda_v)$ and $D_W = \text{diag}(\sqrt{W_v})$ on one of the two relevant allowed-site sets. With the sign suppressed, its off-diagonal matrix is $C_i = D_W T D_W$, restricted to this set. The perturbation of this matrix equals

$$\zeta(D_\lambda C_i + C_i D_\lambda) + \zeta^2 D_\lambda C_i D_\lambda.$$

In the first row term only band rows and their graph neighbors occur. The row weights are at most b_2 , and these neighboring column weights are at most Kb_2 . Factoring the diagonal weights on those row and column sets and using $\|T\| \leq 1$ gives $\|D_\lambda C_i\| \leq \sqrt{K}b_2$. The same argument applies to $C_i D_\lambda$. In the quadratic term both indices lie in the band, giving norm at most b_2 . The estimates apply to every principal restriction. Consequently the full matrix perturbation has norm at most

$$2|\zeta|\sqrt{K}b_2 + |\zeta|^2 b_2. \quad (5.10)$$

Choose a disk of positive radius depending only on y, b_2, K on which Equation (5.10) is less than $y/2$. The imaginary parts of all perturbed pair matrices are then bounded above by $(y/2)I$. With diagonal variable still equal to z , the non-Hermitian version of Lemma 4.2 shows that the

quotient $P_I/P_{I \setminus \{u\}}$ has positive imaginary part. Once again its logarithm has imaginary part in $(0, \pi)$, and the derivative estimate is uniform in the upper weight bound H_0 .

The derivative of $\log P_I$ in this perturbation is

$$\frac{2}{r} \sum_{v \in I} \lambda_v J_i(v, z). \quad (5.11)$$

For completeness, consider the permutation expansion of an outcome determinant. At a target vertex v whose selected color is i , a nontrivial permutation cycle through v uses one row factor and one column factor, hence has derivative multiplier $2\lambda_v$. A fixed point contributes $z + a_v$ and is unchanged because T has zero diagonal. Conditional on color i at v , the sum of the nontrivial terms through v is

$$P_{I \setminus \{v\}}(z)(F_{v,i}^0(z) - z - a_v).$$

The probability of that color at an unassigned vertex is $1/r$. Dividing by P_I proves Equation (5.11), including the contribution of both signs. The same formula holds after deleting u . Apply the derivative bound to the logarithm of the quotient and choose the directions λ_v as before. We obtain

$$\sum_{v \in I \cap \mathcal{U} \cap \{c=0\}} |J_i^I(v, z) - J_i^{I \setminus \{u\}}(v, z)| \leq C_1(y, b_2, K, r). \quad (5.12)$$

The constants in Equations (5.9) and (5.12) can be chosen uniformly for z in any fixed upper-half-plane compact set.

Passage to bounded infinite graphs. Exhaust the graph component of u by finite principal sets. The transforms converge by Lemma 5.1, including the conditional transforms in Equation (5.7). Fatou's Lemma therefore passes both single-deletion estimates to the infinite graph. Their sums can be taken over the entire relation class: there is no effect on other graph components. Moreover, the bounds hold in every comparison environment formed by deleting other vertices first.

The single-deletion estimates are now uniform in the largest weight. It remains to convert them into a comparison for a measurable set of deletions. Directly summing pointwise changes in an arbitrary infinite order would not justify that step. We instead establish the integrated identity first for the local Laurent coefficients and then for the analytic transforms.

The analytic integral change identity. Fix measurable complex directions λ_v , $|\lambda_v| \leq 1$, supported on the band targets: on $\mathcal{U}$ for G , and on $\mathcal{U} \cap \{c=0\}$ for J_i . Let $Z_v(z)$ denote the respective field, extended by zero off those targets. Order E_{hi} measurably and injectively along each class. At a deletion vertex u , let $\delta_u Z_v$ mean presence-minus-absence at u , with earlier vertices already deleted and later vertices still present. In the following formula orient δ_{tot} as before-minus-after. We claim that

$$\int_X \lambda_v \delta_{\text{tot}} Z_v(z) d\nu(v) = \int_{E_{\text{hi}}} \left(\sum_{v \in [u]_{\mathcal{R}}} \lambda_v \delta_u Z_v(z) \right) d\nu(u). \quad (5.13)$$

The left side contains one rooted term, whereas the right side sums over targets in the class of the deleted vertex.

To prove the identity, initially work near infinity in the upper half-plane. Every target is present in both configurations, since the deleted set is disjoint from the band and initially absent band vertices are excluded. At such a target, the bounded spectral support gives

$$G_v(z) = z^{-1} \sum_{k \geq 0} h_k(v) z^{-k}, \quad |h_k(v)| \leq L^k.$$

At a J_i target, the corresponding expansion is

$$F_{v,i}^0(z) - z - a_v = - \sum_{k \geq 0} b_k(v, i) z^{-k-1}, \quad |b_k(v, i)| \leq B_{v,i} L^k,$$

where $B_{v,i}$ is the total mass of the limiting residue measure. These bounds hold because the finite quotients are the negative Cauchy transforms of their residue measures, all supported in $[-L, L]$. The total residue mass is uniformly bounded in this bounded setting by Lemma 4.5. These expansions persist in the limit: each coefficient is eventually fixed by a finite neighborhood, the coefficient bounds are uniform, and the series converge to the limiting transforms for sufficiently large $|z|$. The product defining J_i therefore also has exponentially bounded Laurent coefficients. The coefficient at each order in either field depends on a neighborhood of radius at most a constant times that order, by Lemma 4.5 and multiplication of the series.

For each individual coefficient the ordered local telescoping argument from Lemma 5.2, followed by Equation (5.2), proves the coefficient version of Equation (5.13). The bounded degree ensures that the number of affected roots at a fixed deletion vertex grows at most exponentially in the coefficient order. Combining this with the coefficient bounds just obtained gives absolute summability of the resulting series for sufficiently large $|z|$. Finite measure then allows the sums and integrals to be interchanged. This proves Equation (5.13) on a nonempty open region of the upper half-plane. The region may depend on H_0 , which causes no problem for the analytic continuation that follows.

Both sides of Equation (5.13) are holomorphic on the full upper half-plane. On the right, the sum over v converges absolutely at each z , by the single-deletion estimates. Its analytic partial sums, for example over increasing neighborhoods, are locally bounded uniformly by those same estimates. Normal-family convergence therefore makes the sum holomorphic. The outer integral is over the finite measure set E_{hi} , and the bounds are uniform on compact sets, so it too is holomorphic. On the left, $|G_v(z)| \leq (\text{Im } z)^{-1}$; the residue bound on the band similarly bounds $J_i(v, z)$ on compact sets. These estimates and the finite band measure justify holomorphy of that integral. The identity theorem now proves Equation (5.13) everywhere in the upper half-plane.

Finally fix z and choose the measurable directions λ_v so that $\lambda_v \delta_{\text{tot}} Z_v(z) = |\delta_{\text{tot}} Z_v(z)|$, with arbitrary value when the difference is zero. These are fixed directions for the just-proved identity; they need not depend analytically on z . Its right side is bounded by the single-deletion constant times $\nu(E_{\text{hi}})$. Apply this argument separately to G and J_i and add the estimates. Their constants were independent of H_0 and of the comparison environments, which proves Equation (5.8) with the stated uniformity. $\square$

The integral change identity controls the total positive cost under color assignments and deletions. The transform comparison gives a different control: the effect of distant large weights on any fixed weight band is bounded by the measure of the deleted vertices. Together these are the two estimates needed to define normalized costs on a nonsingular base by passage through invariant restrictions of finite measure.

6 Coloring a nonsingular measured graph

The polynomial calculations of Sections 4 and 5 will now produce a measurable coloring on a probability extension of a nonsingular measured graph. The conclusion concerns the operator norm on a prescribed finite ball, with an arbitrarily small probability of failure. We will also preserve the uniform distribution of each color conditional on the original base point. This latter condition will be needed when the coloring is used inside a von Neumann algebra.

There are two steps in passing from the preceding invariant calculations to this conclusion. We first construct costs on the nonsingular probability space by adding a real height coordinate and then removing upper and lower weight cutoffs. We then assign colors at a small random fraction of the remaining vertices at each step. An estimate for the integrated difference of the limiting costs, on the set where two colorings agree on a large ball, will control the interaction between these simultaneous assignments.

6.1 The probability extension and the coloring theorem

Let (X, μ) be a standard probability space, and let $\mathcal{R}$ be a countable nonsingular measured equivalence relation on X . We use Borel representatives after discarding null sets when necessary. Write the Radon–Nikodym cocycle in the orientation

$$\mu(\gamma S) = \int_S \Delta(\gamma v, v) d\mu(v) \quad (6.1)$$

for measurable partial isomorphisms γ with graph in $\mathcal{R}$ and measurable sets S in their domains. Versions can be chosen off a saturated null set so that the cocycle identity holds. For example, one first uses countably many partial isomorphisms covering $\mathcal{R}$; the chain rule gives the identity for their compositions, and derivatives agree on sets where two such maps coincide. Throughout this Section, null sets arising along the countable relation may be discarded together with their saturations, which are null by nonsingularity.

Let G be an undirected measurable graph contained in $\mathcal{R}$, without loops, and with degree at most $D_0 < \infty$. Its radius- m ball at v is denoted by $B_G(v, m)$. Suppose that

$$K^{-1} \leq \Delta(u, v) \leq K \quad \text{on every edge of } G, \quad K \geq 1. \quad (6.2)$$

We consider a measurable field of Hermitian matrices T on the orbit spaces $\ell^2([v]_{\mathcal{R}})$, with norm at most 1, zero diagonal, and entries supported on G . Upon changing the root within an orbit, the matrices are identified up to conjugacy by a diagonal unitary. All quantities used below, including principal norms and the polynomial costs, are invariant under these conjugacies.

The *relative Bernoulli extension* of $(X, \mu, \mathcal{R})$ is the space

$$\tilde{X} = \{(v, \omega) : v \in X, \omega \in [0, 1]^{[v]_{\mathcal{R}}}\}, \quad \pi(v, \omega) = v,$$

equipped with the probability measure $\tilde{\mu}$ whose conditional measure over v is the product of Lebesgue measures, denoted by κ_v . The equivalence relation on this extension identifies (v, ω) with (u, ω) whenever $(u, v) \in \mathcal{R}$, regarding ω as one function on the common class. Each extended orbit thus projects bijectively onto a base orbit. The graph and matrices are pulled back along this identification.

These are standard measurable spaces and fields: enumerate the base relation by countably many partial isomorphisms, and use one coordinate for each distinct site in a class. Changing the root reindexes a product family and preserves its conditional product measure. Consequently the lift of a base partial isomorphism has the same Radon–Nikodym derivative as that base map. These lifts enumerate the extended relation. Cutting their domains by extension-measurable sets and taking countable disjoint unions therefore shows, by conditional integration, that the cocycle on the extension is the pullback of Δ . A single Lebesgue coordinate at a site will be used to encode countably many independent random coordinates whenever needed.

For an integer $r \geq 1$ and a coloring $c : \tilde{X} \rightarrow \{1, \dots, r\}$, its value at a site u in the orbit of (v, ω) means $c(u, \omega)$. On a finite principal set I in this orbit, let

$$T_{I,c} = \sum_{i=1}^r 1_{\{c=i\} \cap I} T|_I 1_{\{c=i\} \cap I}$$

be the matrix obtained by retaining entries joining vertices of the same color.

Theorem 6.1 (Bernoulli coloring). *Let $(X, \mu, \mathcal{R})$, G , Δ , and T satisfy the assumptions above: $\mathcal{R}$ is countable and nonsingular on a standard probability space, G is undirected, has no loops, has degree at most $D_0 < \infty$, and satisfies Equation (6.2), and T is a measurable Hermitian contraction with zero diagonal and support on G , allowing diagonal unitary conjugacy under a change of root. Fix an integer $r \geq 1$ and a real number $t > 3/\sqrt{r}$, and put $\beta = 80$.*

For every integer $m \geq 1$ and every $\alpha, \eta > 0$, there is a measurable coloring $c : \tilde{X} \rightarrow \{1, \dots, r\}$ on the relative Bernoulli extension such that

$$\kappa_v\{\omega : c(v, \omega) = i\} = \frac{1}{r} \quad \text{for every } i \in \{1, \dots, r\} \quad \text{and almost every } v \in X, \quad (6.3)$$

and

$$\tilde{\mu}\{(v, \omega) : \|T_{B_G(v,m),c}\| \leq 2\beta t + \alpha\} > 1 - \eta. \quad (6.4)$$

We prove the theorem after constructing the cost that will control the assignments. Until the final proof, the probability base may already include some Bernoulli coordinates. We continue to denote this current base by (X, μ) and its relation by $\mathcal{R}$; the cocycle and all graph and matrix bounds remain as above.

6.2 Homogeneous costs from an invariant lift

A partial coloring is a measurable function $c : X \rightarrow \{0, 1, \dots, r\}$, with 0 indicating an unassigned site. Recall the finite-site data of Section 4: the diagonal shift at a site is

$$\tau_v = \begin{cases} t, & c(v) = 0, \\ \beta t, & c(v) > 0, \end{cases}$$

and the allowed polynomial choices are both signs of all r colors at an unassigned site, or both signs of its assigned color otherwise. Choices are uniform among the allowed pairs and independent across sites. With positive weights W_v , the outcome matrix on a choice block is the corresponding principal restriction of

$$\text{diag}(\sqrt{W})(\pm T) \text{diag}(\sqrt{W}) - \text{diag}(\tau W).$$

For the expected characteristic polynomial P and its single-site deletion $Q = P_{-v}$, we use the notation of that Section:

$$F_v = P/Q, \quad G_v = 1/F_v, \quad m_v = \int x_+ \rho_v(dx), \quad d_v = s(P) - s(Q),$$

where ρ_v is the rooted probability measure with Cauchy transform G_v , $x_+ = \max(x, 0)$, and $s(P)$ is the sum of the positive parts of the roots, counted with multiplicity. At an unassigned site the superscript i denotes permanent assignment to color i , including the increase in the shift. The quotient $F_{v,i}^0$ denotes conditioning on color i without increasing that shift, whereas

$$F_{v,i} = F_{v,i}^0 + (\beta - 1)tW_v$$

includes the increase. We set the virtual fields d_v^i to zero on already assigned sites. On bounded invariant measured graphs the same notation refers to the limits supplied by Lemma 5.1.

The original measure need not be invariant, so the integral change formula cannot yet be applied there. We use the Maharam invariant lift [Mah64]; for the multiplicative-coordinate formula, see [RS06, Section 2, before Theorem 2.1]. Here we use logarithmic height, with multiplicative coordinate $y = e^s$; the cocycle verification and subsequent homogeneous-cost construction are given below. Adjoin a height $s \in \mathbb{R}$ and set

$$d\nu(v, s) = d\mu(v)e^s ds, \quad W(v, s) = e^{-s}. \quad (6.5)$$

In the lifted relation, the site corresponding to u at a root (v, s) is

$$(u, s - \log \Delta(u, v)). \quad (6.6)$$

The measure ν is invariant. Indeed, the factor $\Delta(u, v)$ from Equation (6.1) cancels the factor $e^{-\log \Delta(u, v)}$ from the height density. The destination weight is $\Delta(u, v)W(v, s)$, so neighboring weights have ratios in $[K^{-1}, K]$. The coloring and matrix data are independent of height.

For $0 < l < H < \infty$, retain only the sites with $l \leq W \leq H$, using the restricted relation, graph, and principal matrix data. The retained measure is finite, since

$$\nu\{l \leq W \leq H\} = l^{-1} - H^{-1}.$$

All results of Section 5 therefore apply. Denote the resulting cost fields by $m^{l,H}, d^{l,H}, (d^i)^{l,H}$. The next lemma removes both cutoffs, while retaining control uniform enough for later changes of coloring.

Lemma 6.2 (Homogeneous costs). *Fix D_0, K, r, t and $\beta = 80$, and let the probability space, relation, graph, matrix data, and partial coloring satisfy the assumptions above. For each field f in the finite list $m, d, d^1, \dots, d^r$, there is a bounded measurable function $\bar{f}$ on X such that, on every fixed band*

$$\mathcal{U}_{b_1, b_2} = \{(v, s) : b_1 \leq W(v, s) \leq b_2\}, \quad 0 < b_1 < b_2 < \infty,$$

one has

$$\frac{f^{l,H}}{W} \longrightarrow \bar{f}(v) \quad \text{in } L^1(\mathcal{U}_{b_1, b_2}, \nu) \quad \text{jointly as } l \downarrow 0, H \uparrow \infty. \quad (6.7)$$

The errors tend to zero uniformly over all such probability bases and configurations with the fixed parameters and bounds. There is a constant C_0 , depending only on those parameters, such that all the normalized cutoff fields on retained sites and all their limits have absolute value at most C_0 . The limit fields are independent of height and satisfy

$$\min_{1 \leq i \leq r} \bar{d}^i - \bar{d} \leq 24\bar{m} \quad \text{almost everywhere on } \{c = 0\}. \quad (6.8)$$

Proof. The two cutoffs use different controls. The set of weights above H has invariant measure H^{-1} , so the rank comparison controls the upper end. With the upper cutoff fixed, the edge-ratio bound makes sufficiently small weights invisible in any chosen finite neighborhood of a fixed band. We first prove uniform convergence on the band $1 \leq W \leq e$. The same argument applies to any fixed band bounded away from zero. Keep the lower cutoff l fixed and compare upper cutoffs $H_2 \geq H_1 \geq e$. Passing from H_2 to H_1 deletes a set of invariant measure at most $1/H_1$. Lemma 5.3 gives, on the appropriate sites of the band, integrated absolute differences of order $1/H_1$ for

$$G_v(z) \quad \text{and} \quad J_i(v, z) = (F_{v,i}^0(z) - z - a_v)G_v(z), \quad a_v = \tau_v W_v.$$

For J_i the targets are the unassigned band sites. The constants are uniform in l, H_1, H_2 , the configuration, and z on each compact subset of the open upper half-plane.

The second-moment estimate of Lemma 4.5 is

$$\int x^2 \rho_v(dx) \leq ((\beta t)^2 + K) W_v^2. \quad (6.9)$$

It also holds for each of the altered site choices used here. In particular, these second moments are uniformly bounded on the band. They give a uniform positive lower bound for $|G_v(z)|$ on each compact subset of the upper half-plane: a fixed bounded interval carries a uniformly positive amount of the probability measure, and

$$-\operatorname{Im} G_v(z) = (\operatorname{Im} z) \int |z - x|^{-2} \rho_v(dx).$$

The residue bound in Lemmas 4.2 and 4.5 also bounds J_i uniformly on these compact sets. Division is thus controlled in the identities

$$F_v = 1/G_v, \quad F_{v,i}^0 = z + a_v + J_i/G_v, \quad F_{v,i} = F_{v,i}^0 + (\beta - 1)tW_v.$$

The integrated differences of these quotients consequently tend to zero as $H_1 \rightarrow \infty$, uniformly in the lower cutoff.

We next recover the costs from these transform comparisons. Probability measures with the uniform second-moment bound form a weakly compact family, and integration of x_+ is continuous on this family by uniform integrability. Cauchy transforms on a countable dense subset of the upper half-plane separate probability measures. Compactness therefore implies that, for any prescribed accuracy of m_v , comparisons of the transforms at finitely many arguments to sufficiently small accuracy suffice. The integrated transform estimates, together with the uniform bound on m_v , give the desired integrated comparison for m .

For deletion costs we use the argument of the quotient. In a finite restriction, let $n_P(x)$ and $n_Q(x)$ count roots strictly greater than x , with multiplicity. Interlacing and the quotient properties give, for $y > 0$,

$$\arg F(x + iy) = \int_{\mathbb{R}} \frac{y}{(x - u)^2 + y^2} (n_P - n_Q)(u) du, \quad 0 \leq \arg F \leq \pi. \quad (6.10)$$

This follows by writing the arguments of the individual root factors; the branch is fixed by $\operatorname{Im} F > 0$. The spectral-shift function $n_P - n_Q$ takes values between zero and one. Lemma 4.4 and Equation (6.9) imply that

$$\int_B^\infty (n_P - n_Q)(x) dx = O(B^{-1}) \quad (B \rightarrow \infty), \quad (6.11)$$

uniformly on the band, including for the shifted virtual options. Indeed, the bound in that lemma is

$$2 \int (x - B)_+ \rho_v(dx) + (a_v + B) \rho_v\{x > B\},$$

whose two terms are controlled by the second moment and the band bound on a_v .

On $[0, B]$, the integral of the spectral-shift function is approximated by

$$\frac{1}{\pi} \int_0^B \arg F(x + iy) dx \quad \text{as } y \downarrow 0, \quad (6.12)$$

uniformly over these finite calculations. To see uniformity, interchange the integrals in Equation (6.10); the error is at most the $L^1(\mathbb{R})$ error in the Poisson approximation to $1_{[0,B]}$, because $0 \leq n_P - n_Q \leq 1$. More precisely, put

$$p_y(u) = \frac{y}{\pi(u^2 + y^2)}, \quad \delta_B(y) = \|p_y * 1_{[0,B]} - 1_{[0,B]}\|_{L^1(\mathbb{R})}.$$

For fixed B , one has $\delta_B(y) \rightarrow 0$ as $y \downarrow 0$. Combining the smoothing error with Equation (6.11) gives, uniformly for the finite rooted calculations on the band,

$$\left| d_v - \frac{1}{\pi} \int_0^B \arg F_v(x + iy) dx \right| \leq \frac{C}{B} + \delta_B(y), \quad B \geq 1. \quad (6.13)$$

The same estimate holds for each shifted virtual option. For fixed B, y , convergence of the costs and locally uniform convergence of the quotients in Lemma 5.1 pass this estimate to every present band vertex of a bounded infinite cut, and to present unassigned band vertices for the virtual options. The quotient has imaginary part at least y , so its argument is continuous throughout this passage. We use this estimate in terms of the cost and quotient on the infinite cut. For fixed B, y , the arguments in its integral are controlled by the integrated quotient comparisons already proved. First choose B large, then y small for that B , and finally the upper cutoff H_1 large for the fixed segment $[0, B] + iy$. This proves the high-cutoff comparison for d and every d^i .

It remains to remove the lower cutoff. For a fixed finite upper cutoff, Lemma 5.1 approximates the costs uniformly by bounded neighborhood calculations. A radius- R neighborhood of a root in the fixed band cannot contain a vertex of weight below K^{-R} . Thus a sufficiently small lower cutoff is invisible to any one of these fixed neighborhood calculations. This proves uniform convergence as $l \downarrow 0$ with that upper cutoff fixed. For joint convergence, first compare both large upper cutoffs to one sufficiently large fixed upper cutoff by the preceding estimates, and then compare the lower cutoffs there. This gives uniform joint L^1 -Cauchy estimates. Division by W does not change convergence on a fixed band.

The uniform bound asserted in the lemma follows directly from Equation (6.9) and Lemma 4.4: m_v/W_v is uniformly bounded, and $d_v \leq 2m_v + a_v$. The same bounds hold for all virtual options. In particular they are independent of the band, the cutoff, and the configuration.

We have constructed limits on every fixed band. To identify their height dependence, translate height by a real number b . This scales every weight by e^{-b} and changes the cutoffs correspondingly. All outcome matrices and all costs scale by the same positive factor. Thus

$$\frac{f^{l,H}(v, s + b)}{W(v, s + b)} = \frac{f^{e^{bl}, e^b H}(v, s)}{W(v, s)}.$$

Joint convergence makes the limiting normalized field invariant almost everywhere under every rational height translation, simultaneously. A measurable function on $\mathbb{R}$ invariant almost everywhere under all rational translations is constant almost everywhere; this follows, for example, by convolution. Applying this on almost every base fiber gives the functions $\bar{f}(v)$ and their asserted independence of height. Their measurability also follows from the band limits, for instance by integrating a representative over a fixed height interval.

Finally Lemma 4.6 gives the single-site inequality on finite restrictions. Lemma 5.1 passes it to each bounded cut. Joint band convergence then permits a pointwise convergent subsequence for the finite list of normalized fields. Dividing by the positive root weight and taking this limit yields Equation (6.8). $\square$

We record two compatibility properties of this construction for later use. Suppose $(\hat{X}, \hat{\mu}, \hat{\mathcal{R}})$ is a standard nonsingular extension with a measurable map $\pi : \hat{X} \rightarrow X$ such that $\pi_* \hat{\mu} = \mu$, each relation class projects bijectively onto an original class, and its Radon–Nikodym cocycle is $\hat{\Delta}(\hat{u}, \hat{v}) = \Delta(\pi(\hat{u}), \pi(\hat{v}))$. Pull back the partial coloring, graph, and matrix data. This includes adjoining independent Bernoulli coordinates that the current coloring does not use. Every bounded-cut ball calculation is then exactly a pullback. Since π preserves the base probability, the corresponding band L^1 error norms agree. Uniqueness of the limits in Lemma 6.2 therefore gives the same pullback identity almost everywhere for each normalized limit field.

For a permutation σ of the color names, extended by $\sigma(0) = 0$, the finite polynomial laws are relabeled by the same permutation. The bounded-cut costs and their unique limits consequently satisfy

$$\bar{m}(\sigma \circ c) = \bar{m}(c), \quad \bar{d}(\sigma \circ c) = \bar{d}(c), \quad \bar{d}^i(\sigma \circ c) = \bar{d}^{\sigma^{-1}(i)}(c) \quad (1 \leq i \leq r)$$

almost everywhere. The first two identities express invariance of the cost, while the last records how the virtual color options are renamed.

Define the nonnegative total cost of a partial coloring on the probability base by

$$\mathcal{S}(c) = \int_X \bar{m}(v) d\mu(v). \quad (6.14)$$

The preceding construction gives a uniform bound for this quantity. The next lemma records the two approximation properties that will allow us to use it: convergence after averaging over a long height strip, and small integrated changes when two colorings agree on a large ball.

Lemma 6.3 (Strip averages and locality). *Under the assumptions of Lemma 6.2, put*

$$\Sigma_n = \{e^{-n} \leq W \leq e^n\}, \quad f_{\Sigma_n} = f^{e^{-n}, e^n}.$$

For every field f in the list $m, d, d^1, \dots, d^r$,

$$\frac{1}{2n} \int_{\Sigma_n} |f_{\Sigma_n} - W\bar{f}| d\nu \longrightarrow 0 \quad (n \rightarrow \infty), \quad (6.15)$$

uniformly over the configurations and probability bases with the fixed parameters.

If c_1, c_2 are two partial colorings on the same base with the same graph, matrix, and cocycle data, and E_R is the set where they agree on the radius- R graph ball, then

$$\int_{E_R} |\bar{f}(c_1) - \bar{f}(c_2)| d\mu \leq \gamma(R), \quad \gamma(R) \longrightarrow 0, \quad (6.16)$$

where the same function γ may be used for the finite list of fields and is uniform over these configurations.

Both assertions hold after taking a product with an auxiliary probability space and keeping the auxiliary parameter fixed along the equivalence relation. The cost fields may be chosen measurably on such product spaces.

Proof. In the strip integral there is the identity

$$|f_{\Sigma_n} - W\bar{f}| d\nu = |f_{\Sigma_n}/W - \bar{f}| d\mu ds.$$

Divide the height interval $[-n, n]$ into unit intervals. Translating each to a fixed interval uses the homogeneity proved in Lemma 6.2. For intervals a sufficiently large fixed distance from both strip ends, the translated lower and upper cutoffs are uniformly close to zero and infinity, respectively.

Uniform joint convergence on the fixed band bounds their average errors by any prescribed positive number. The remaining bounded number of end intervals has uniformly bounded error, and its contribution divided by $2n$ tends to zero. This proves Equation (6.15).

For Equation (6.16), first approximate both limiting normalized fields on a fixed unit height band in L^1 by common, fixed lower and upper cutoffs. The errors are uniform by Lemma 6.2. On this bounded cut, Lemma 5.1 approximates each field uniformly by a finite-radius calculation. Agreement of the base colorings on a sufficiently large ball makes these lifted local data identical. Since the normalized limit fields are independent of height, integration on the unit band bounds the required base integral. Choosing the cutoff and then the radius for each prescribed accuracy gives Equation (6.16).

Finally, all bounded-cut calculations are measurable, being limits of rooted finite calculations. They remain so with additional measurable parameters. An auxiliary parameter held fixed along the relation does not alter the degree, edge ratios, or any spectral estimate. The preceding uniform estimates therefore hold on its product probability space as well, and the L^1 limit construction gives jointly measurable limit fields there. $\square$

6.3 The cost detects balls with excessive norm

We next relate $\mathcal{S}$ to the norm conclusion in Theorem 6.1. The all-unassigned configuration has zero cost. Once a ball is fully assigned, a violation of the desired norm bound forces a definite positive-root cost in that ball. The invariant lift allows us to sum this contribution without assuming that the original probability measure is invariant.

Lemma 6.4 (Initial cost and bad balls). *Assume the hypotheses and parameter choices of Theorem 6.1. The all-unassigned partial coloring has $\mathcal{S} = 0$. Fix $m \geq 1$ and $\alpha > 0$. For any measurable partial coloring c , let $\text{Bad}_m(c)$ be the set of vertices v for which every vertex of $B_G(v, m)$ is assigned and*

$$\|T_{B_G(v, m), c}\| > 2\beta t + \alpha.$$

There is a finite constant C_4 , depending only on m, D_0, K, α , such that

$$\mu(\text{Bad}_m(c)) \leq C_4 \mathcal{S}(c). \quad (6.17)$$

The same assertions hold on probability extensions with the pulled-back graph, matrix, and cocycle data.

Proof. For the all-unassigned coloring, Lemma 4.3 gives zero cost on every finite principal restriction. Finite restriction limits and Lemma 6.2 therefore give $\mathcal{S} = 0$.

To prove Equation (6.17), partition the bad centers into a bounded finite number of measurable classes so that two distinct centers in one class have graph distance greater than $2m + 1$. Here is a measurable construction. A bounded-degree Borel graph has a countable measurable partition into independent sets: assign to each vertex a finite prefix of an injective standard-space code long enough to separate it from all of its neighbors, and include the prefix length in the assigned code. Equal assigned codes cannot occur at adjacent vertices. Visit these independent sets in sequence, and give each vertex the first color not already used by one of its previously colored neighbors, from a palette of one more than the degree bound. This is a measurable proper coloring. Apply this construction to the graph whose edges join vertices at distance at most $2m + 1$, and intersect its color classes with $\text{Bad}_m(c)$. The number of resulting classes is bounded in terms of D_0, m .

Fix one such class C of bad centers. In the invariant lift retain the strip Σ_n , and select the lifted centers above C whose height is at distance at least $m \log K$ from both ends of $[-n, n]$. Every

radius- m ball around a selected center lies entirely in the strip. Moreover these balls are disjoint and nonadjacent along each lifted orbit. Indeed Equation (6.6) and the cocycle law give exactly one lifted site above each base site in a fixed lifted orbit. Distinct centers there therefore project to distinct centers of C , whose distance is greater than $2m + 1$.

Delete every strip site outside these balls. By Lemma 5.2, deletion cannot increase the total integrated cost, so

$$\int_{\Sigma_n} m_{\Sigma_n} d\nu \geq \int m_{\text{retained}} d\nu. \quad (6.18)$$

Each retained graph component is one of the selected finite balls. Invariant mass transport sends the rooted costs in each such ball to its center, and hence the right side of Equation (6.18) equals the integral over the selected lifted centers of $s(P_{\text{ball}})$.

For a bad ball, the second part of Lemma 4.3 gives $s(P_{\text{ball}}) \geq (\alpha/2) \min_{\text{ball}} W$. The edge-ratio bound then gives

$$s(P_{\text{ball}}) \geq \frac{\alpha}{2} K^{-m} W_{\text{center}}. \quad (6.19)$$

Divide Equation (6.18) by $2n$ and let $n \rightarrow \infty$. Equation (6.15) makes the left side tend to $\mathcal{S}(c)$. On the right, $W d\nu = d\mu ds$, and the allowed height interval has length $2n - 2m \log K$ for all sufficiently large n . Thus Equation (6.19) gives

$$\frac{\alpha}{2} K^{-m} \mu(C) \leq \mathcal{S}(c).$$

Summing over the bounded number of classes proves Equation (6.17). Every part of the argument uses only the stated probability-space, cocycle, degree, and matrix bounds, so it applies unchanged to the indicated extensions. $\square$

6.4 Assigning a small fraction of the remaining vertices

We now use Equation (6.8) to choose colors. Changing one site has a controlled cost, but simultaneous changes must be compared in their intermediate environments. Independent uniform times provide an order for those comparisons. We first perform the comparison on a bounded height strip, where the costs have deterministic definitions by finite-neighborhood limits. Only then do we pass to the homogeneous costs. This order of operations will avoid evaluating a version of an L^1 limit at a random value of its parameter.

Fix a current partial coloring c determined by coordinates already used, and let $E_0 = \{c = 0\}$. At every site in E_0 , choose a color $i(v)$ minimizing $\bar{d}^i(v)$. Break ties by a fresh uniform random permutation of the colors, independently at each site, taking the first minimizer in that permutation. The costs of c are first defined on the space carrying the old coordinates and then pulled back to the space carrying the extra tie coordinates. The bounded-cut definitions and their limits commute with this pullback. In particular, the choices $i(v)$ use only old and tie coordinates.

Now add fresh independent uniform times $U_v \in (0, 1)$ at the sites, independent of both the old and the tie coordinates. For $0 \leq \theta < 1$, define c^θ by assigning $i(v)$ at each site $v \in E_0$ with $U_v < \theta$ and leaving all other sites as in c . All these definitions are measurable on the relative Bernoulli extension. Null exceptions can be filled by measurable default values before adding the next independent coordinates; their saturations are negligible. We use this convention throughout the finite sequence of steps below.

Lemma 6.5 (One assignment step). *Under the hypotheses of Theorem 6.1, let c be any partial coloring determined before the fresh times, and choose $i(v)$ and c^θ as just described. There is a*

nonnegative function g , depending only on the fixed graph and polynomial parameters, such that $g(h) \rightarrow 0$ as $h \downarrow 0$ and, for every $0 < h < 1/2$,

$$\mathcal{S}(c^h) \leq (1 + 24h)\mathcal{S}(c) + hg(h). \quad (6.20)$$

The same function g works for all such steps and current configurations.

Proof. Apply Lemma 5.2 to the change from c to c^h on the invariant lift restricted to Σ_n . Order the changed sites by their times U_v . These times are almost surely distinct along every countable orbit, after discarding a saturated null set. The order need not be a well-order, as allowed in that lemma. At a comparison site v the pre-change configuration is the restriction of c^{U_v} , with the number U_v held fixed as a threshold along that orbit.

We claim that the resulting integral of the single-site increments is

$$\begin{aligned} & \int_{\Sigma_n} (m_{\Sigma_n}(c^h) - m_{\Sigma_n}(c)) d\nu \\ &= \int_0^h \frac{d\theta}{1-\theta} \int_{\Sigma_n} 1_{\{v \in E_0, U_v \geq \theta\}} \left[d_{\Sigma_n}^{i(v)}(c^\theta) - d_{\Sigma_n}(c^\theta) \right] d\nu. \end{aligned} \quad (6.21)$$

Here ν includes all label laws as well as the height measure, and v denotes the current base site of a lifted root.

To justify the claim, condition at a root in E_0 on the old coordinates, all tie coordinates, and every fresh time except the time at that root. Hold the height fixed as well. For a fixed threshold θ , leave the root site unassigned, and denote the bracketed difference in Equation (6.21) by $D(\theta)$ for these fixed other data. Whenever the root time u_0 is at least θ , this difference is independent of its actual value. The chosen colors were determined before the fresh times, other sites use only their own fresh times in the threshold test, and neither the graph nor the restricted weight data use this new root coordinate. This independence holds for the deterministic rooted finite-subset calculations and hence for the bounded-strip costs obtained from them. The same description applies at $u_0 = \theta$ because the update condition is strict.

When $u_0 < h$, the ordered comparison increment at the root is therefore $D(u_0)$, outside the null event of a tie with another site. Integrating over this last uniform time uses precisely the identity

$$\int_0^h D(u_0) du_0 = \int_0^h \frac{d\theta}{1-\theta} \int_\theta^1 D(\theta) du_0.$$

The bounded-strip computations are measurable jointly in the threshold and in the remaining data, again by their finite rooted definitions. Fubini's Theorem, first for almost every root and height and then over the remaining coordinates, proves Equation (6.21). This argument takes place entirely before choosing any homogeneous limit version depending on θ .

We can now pass to that limit in an integrated expression. Divide Equation (6.21) by $2n$ and use Lemma 6.3. For the right side, adjoin θ with normalized Lebesgue probability on $[0, h]$, holding it fixed along the equivalence relation, and regard c^θ as one configuration on this augmented base. The lemma applies uniformly there. Its absolute integrated errors also control the errors after multiplication by the displayed mask, after selection among the finite list of options $i(v)$, and after multiplication by $(1-\theta)^{-1} \leq 2$. For a height-independent limit field, cancellation of W against the height density leaves a height interval of length exactly $2n$. Consequently

$$\mathcal{S}(c^h) - \mathcal{S}(c) = \int_0^h \frac{d\theta}{1-\theta} \int_X 1_{\{v \in E_0, U_v \geq \theta\}} \left[\bar{d}^{i(v)}(c^\theta) - \bar{d}(c^\theta) \right] d\mu. \quad (6.22)$$

All fields in this formula have jointly measurable versions on the augmented space. No evaluation of an unspecified limit at the random diagonal $\theta = U_v$ has been used.

It remains to compare the intermediate costs with the old costs, for which $i(v)$ was chosen. Let D_R be a finite bound, depending only on D_0, R , for the cardinality of a radius- R graph ball. On the event that this ball has no update from c to c^θ , the two partial colorings agree there. Equation (6.16) controls the integrated difference of their cost fields on that event; summing over the finite option list also controls the selected field. Conditional on the old and tie coordinates, the probability of an update anywhere in the ball is at most $D_R h$, by the union bound for the fresh uniform times. On the event that an update does occur, the cost differences are uniformly bounded by Lemma 6.2.

These observations apply either for each fixed θ , or jointly with normalized measure on $[0, h]$ as above. They show that replacing the two costs in the brackets of Equation (6.22) by their values at c creates an error in absolute value at most

$$Ch(\gamma(R) + D_R h), \quad (6.23)$$

where C depends only on the fixed parameters. The factor $(1 - \theta)^{-1}$ is bounded by 2.

The old costs and the minimizing choice $i(v)$ are independent of the fresh root time. The survival probability $1 - \theta$ therefore cancels the denominator in Equation (6.22). By Equation (6.8), the resulting old-cost contribution is at most

$$h \int_{E_0} (\bar{d}^{i(v)}(c) - \bar{d}(c)) d\mu(v) \leq 24h \int_{E_0} \bar{m}(c) d\mu \leq 24h\mathcal{S}(c).$$

For each prescribed accuracy in Equation (6.23), first take R large enough to control $\gamma(R)$ and then take h small enough to control $D_R h$. Equivalently, taking the infimum of these uniform error bounds over integer R gives a nonnegative function $g(h) \rightarrow 0$. This proves Equation (6.20) uniformly over the current configurations. $\square$

6.5 Completion of the coloring

The estimates now have the two features needed for a finite construction. Repeated small assignment steps can cover any prescribed ball with high probability, and, for a fixed total assignment time, their accumulated cost tends to zero with the step size. The bad-ball estimate then supplies the norm conclusion.

Proof of Theorem 6.1. Work on the relative Bernoulli extension of the original base, reserving independent coordinates at every site for each step. Start with the all-unassigned coloring, whose cost is zero by Lemma 6.4. Choose $T_0 > 0$ and $h \in (0, 1/2)$, and apply the construction of Lemma 6.5 with independent fresh coordinates for

$$N = \lceil T_0/h \rceil$$

steps. Write S_j for the cost after j steps. Iterating Equation (6.20) gives

$$S_N \leq \frac{g(h)}{24} ((1 + 24h)^N - 1). \quad (6.24)$$

For fixed T_0 this tends to zero as $h \downarrow 0$, since $(1 + 24h)^N \leq \exp(24(T_0 + h))$.

An unassigned site receives a color at its first successful hazard. Conditional on the original base point, its probability of remaining unassigned after all N steps is $(1 - h)^N \leq e^{-T_0}$. This conclusion does not depend on the color selected at a successful step. If D_m bounds the size of a radius- m ball,

the conditional probability that any site in that ball remains unassigned is at most $D_m e^{-T_0}$. The unconditional probability that the ball is fully assigned but violates the required norm bound is at most $C_4 S_N$, by Lemma 6.4. First choose T_0 large, and then h small, so that

$$D_m e^{-T_0} + C_4 S_N < \eta.$$

Complete the partial coloring by fresh independent uniform colors at the remaining sites. Every previously complete ball is unchanged. The resulting coloring therefore satisfies Equation (6.4).

It remains to verify the conditional marginals. Fix a permutation σ of $\{1, \dots, r\}$, and let Θ_σ apply it to the color names in every tie-breaking permutation and every final uniform color input, leaving all times unchanged. This transformation fixes the original base point, commutes with rerooting, and preserves each conditional product measure κ_v . On the height lift it also fixes the height, so it acts isometrically on every band L^1 space.

Suppose inductively that the current coloring satisfies $c \circ \Theta_\sigma = \sigma \circ c$. Exact pullback of the finite-cut calculations and their color covariance give

$$((d^{\sigma(i)})^{l,H}(c)) \circ \Theta_\sigma = (d^i)^{l,H}(c), \quad 1 \leq i \leq r.$$

The band L^1 isometry and uniqueness of the homogeneous limits give $(\bar{d}^{\sigma(i)}(c)) \circ \Theta_\sigma = \bar{d}^i(c)$ almost everywhere. Thus the set of minimizing colors at the transformed input is the σ -image of the original minimizing set. The transformed permutation tie-breaker chooses the corresponding color, and the update test is unchanged because the times are unchanged.

There are only finitely many color permutations and, with T_0, h fixed, finitely many steps. Choose a common conull set for the required identities, intersect its images under the finite color action, and discard the saturation of its complement under the relation. The induction just described then holds throughout the construction on that set, starting from the all-unassigned coloring. The final uniform color inputs transform in the same way, so the completed coloring also obeys $c \circ \Theta_\sigma = \sigma \circ c$.

Since Θ_σ preserves the conditional product measure over each original base point, disintegration shows that all output colors at the root have equal conditional probabilities for almost every base point. They sum to one, which proves Equation (6.3) and completes the proof. $\square$

7 A Bernoulli model and transfer of state moments

Throughout this Section, $A \subset M$ is a maximal abelian subalgebra, M has separable predual, and φ is a faithful normal state with $A \subset M_\varphi$. We retain the notation of Section 3: N is the algebra generated by the groupoid normalizers of A , $E_N : M \rightarrow N$ is the φ -preserving conditional expectation, $M^\circ = \ker E_N$, and N_{alg} is the algebra of finite sums of base functions times kernel groupoid normalizers. The Cartan inclusion $A \subset N$ is represented by a countable nonsingular measured equivalence relation $\mathcal{R}$ on (X, μ) , with its scalar cocycle.

The coloring constructed in Section 6 belongs to the Bernoulli extension of this relation. We first place that extension and M in a common von Neumann algebra. The resulting model gives a norm bound for the part of an operator in M° . We then show that every fixed scalar moment involving finitely many model cylinder functions is a limit of averaged moments in M . This is the step that will bring the model color projections back into A .

7.1 The model and a free compression bound

Let $\tilde{X}$ be the relative Bernoulli extension of $(X, \mu, \mathcal{R})$ from Section 6, with one Lebesgue label at each site of a relation class. Write $\tilde{\mu}$ for its probability measure and $B = L^\infty(\tilde{X}, \tilde{\mu})$. Re-rooting

preserves the label assignment as a function on the sites. In particular, each extended relation class projects bijectively onto its base class. Let $\tilde{N}$ be the von Neumann algebra of the extended relation, with the scalar cocycle of N pulled back to the extension.

Lemma 7.1 (The Bernoulli free-product model). *The pullback of base kernel operators identifies N with a von Neumann subalgebra of $\tilde{N}$. There is a faithful normal conditional expectation*

$$\tilde{E}_N : \tilde{N} \longrightarrow N$$

whose restriction to B integrates labels conditional on the base point. If $\tilde{\varphi}$ denotes integration of the diagonal in $\tilde{N}$, then

$$(\varphi|_N) \circ \tilde{E}_N = \tilde{\varphi}.$$

Moreover, the reduced amalgamated free product

$$\mathcal{K} = M *_N \tilde{N}$$

has a faithful normal conditional expectation $E_N^{\mathcal{K}}$ onto N , restricting to E_N and $\tilde{E}_N$. The state

$$\Phi = (\varphi|_N) \circ E_N^{\mathcal{K}}$$

is faithful and normal, restricts to φ and $\tilde{\varphi}$ on the two constituent algebras, and satisfies $B \subset \mathcal{K}_\Phi$. The two constituent algebras are free over N : the expectation of every alternating product of letters from $\ker E_N$ and $\ker \tilde{E}_N$ is zero.

Proof. Use the right-counting direct-integral representations recalled in Section 3. Under the bijection between extended and base classes, a pulled-back base kernel acts by the same orbit operator on each fiber. Pullback is a normal faithful representation: it is an amplification over the conditional probability spaces of labels. Equivalently, matrix coefficients against vector fields on the extension disintegrate to normal functionals on the original decomposable algebra. The pulled-back copy of N lies in $\tilde{N}$, since this holds for the generating kernel normalizers.

The isometry from the original representation space to the extension space that makes a vector field constant in the labels intertwines these copies of N and sends the original diagonal state vector to the new one. Compression by this isometry maps $\tilde{N}$ normally into N . To see the range assertion, first consider finite sums of B -functions times lifted base groupoid normalizers. Compression integrates the coefficient functions over the labels at the relevant sites and hence has its range in N . These finite sums form a weakly dense $*$ -algebra: the lifted partial isomorphisms enumerate the extended relation, and their cuts by extension-measurable sets, together with scalar phase functions, give its kernel normalizers. Normality of compression and weak closure of N give the range assertion for all of $\tilde{N}$.

The compression is unital, N -bimodular, and equal to the identity on N , so it gives the asserted normal conditional expectation. It preserves the diagonal state. That state is faithful by the faithful normal diagonal expectation for a countable nonsingular relation algebra. Consequently $\tilde{E}_N$ is faithful as well. The diagonal expectation is B -bimodular and B is abelian, so $B \subset \tilde{N}_{\tilde{\varphi}}$.

Both expectations onto N are faithful and normal, and the constituent algebras are σ -finite. The von Neumann algebra reduced amalgamated free-product construction therefore gives $\mathcal{K}$, $E_N^{\mathcal{K}}$, generation by the two copies, and their freeness over N ; see [Ued99, Section 2] and [Ued13, Section 2]. These hypotheses are the faithful-normal conditional-expectation hypotheses of that construction, and do not require tracial states. Faithfulness and normality of Φ follow from those of the expectation and of $\varphi|_N$.

For completeness, the product has Φ -preserving conditional expectations onto each constituent algebra. Fix one constituent and compress the state representation of $\mathcal{K}$ to the closure of its state vectors. The linear span of N and alternating centered words is weakly dense. Every reduced word containing a centered letter from the other constituent compresses to zero: test between vectors from the retained algebra, expand at the two ends, and absorb the resulting N -coefficients into adjacent letters; freeness then applies. The retained algebra acts normally and faithfully on its state-vector space. Compression is thus a normal bimodular state-preserving map onto that represented algebra. By Takesaki's Theorem [Tak72], the modular group of Φ restricts to the modular group of each constituent. This also agrees with [Ued13, Equation (2.1)]. Since B is in the centralizer of $\tilde{\varphi}$, it is in $\mathcal{K}_\Phi$. $\square$

Lemma 7.2 (Free compression). *In the model of Lemma 7.1, let $p \in B$ be a projection such that $\tilde{E}_N(p) = \lambda 1$, where $0 \leq \lambda \leq 1$. For every $w \in M^\circ$,*

$$\|pwp\| \leq (\lambda + 2\sqrt{\lambda})\|w\|. \quad (7.1)$$

Proof. In $L^2(\mathcal{K}, \Phi)$, let P be the orthogonal projection onto the closed span of reduced-word vectors whose first letter is a centered M -letter. The orthogonal complement is spanned densely by the N -vectors and the reduced-word vectors whose first letter is a centered $\tilde{N}$ -letter. Both the spanning assertion and the orthogonality follow from generation and freeness over N .

Left multiplication by w sends each of these generators of the orthogonal complement into the range of P . Thus

$$(1 - P)w(1 - P) = 0.$$

On a word beginning with a centered M -letter, left multiplication by $p - \lambda 1$ inserts a centered $\tilde{N}$ -letter at the beginning. Its image is orthogonal to the range of P . Hence

$$PpP = \lambda P, \quad \|Pp\|^2 = \|PpP\| \leq \lambda.$$

Insert $P + (1 - P)$ on both sides of w in pwp . The term with $(1 - P)w(1 - P)$ vanishes. The remaining three terms have norms at most $\lambda\|w\|$, $\sqrt{\lambda}\|w\|$, and $\sqrt{\lambda}\|w\|$, respectively. This proves Equation (7.1). $\square$

7.2 Finite symbols and their evaluation in the masa

We now specify the model functions that can be evaluated inside A . Fix an integer $s \geq 2$, and partition each Lebesgue label into s equally likely symbols, identified with the s th roots of unity. A finite dyadic symbol partition, with s a power of two, will suffice when we later approximate arbitrary labels.

A *finite cylinder function* is a bounded function on $\tilde{X}$ that depends measurably on the base point and on these symbols at a fixed finite list of translated sites. Each translated site is specified by a partial isomorphism with graph in $\mathcal{R}$; the function may also depend on whether the base point belongs to that map's domain. Let $\mathcal{Y}$ be the algebra of finite sums of such functions times lifted base kernel groupoid normalizers. Covariance and composition show that this is a $*$ -algebra. Conditional label integration gives

$$\tilde{E}_N(\mathcal{Y}) \subset N_{\text{alg}}.$$

Choose increasing finite Borel partitions $\mathcal{P}_j$ of X that generate A and separate points outside the fixed null exceptions. For one such partition write $\mathcal{P} = \{e_a\}_a$, identifying its pieces with their projections in A . For independent uniform s th roots of unity z_a , define

$$u_{\mathcal{P}} = \sum_a z_a e_a \in A.$$

Replace the symbol at a translated site by the value of $u_{\mathcal{P}}$ at that site, and replace every lifted base normalizer by its original normalizer in N . This defines an evaluation map, denoted by $\text{ev}_{\mathcal{P}}: \mathcal{Y} \rightarrow N_{\text{alg}}$. Finite cylinder functions themselves evaluate to elements of A .

This evaluation respects multiplication and adjoints and fixes N_{alg} . Substitution is consistent under re-rooting and therefore with covariance. It also respects almost-everywhere identities. Indeed, conditional on almost every base point, each assignment of symbols to any finite list of distinct sites has positive probability. An identity of cylinder functions therefore holds for every such assignment outside a null set of base points, including the assignments produced by $u_{\mathcal{P}}$. The same reasoning applies to identities between finite sums of cylinder coefficients times normalizers: check the finitely supported root columns in the counting representations, including the common cocycle phases. Equality of these columns almost everywhere gives equality of the operators by faithfulness of the diagonal state. In particular, the definition does not depend on the chosen finite expression.

The norm of an evaluated cylinder coefficient is bounded by its original essential bound. For every fixed finite normalizer sum, its expression consequently gives a uniform bound on all evaluations. No uniform bound over arbitrary finite expressions will be needed.

Before stating the transfer result, we isolate the uniform approximation estimate needed for its fixed scalar words. It holds for any faithful normal state and will also be used in the final clipping argument.

Lemma 7.3 (Uniform right multiplication in the state norm). *Let φ be a faithful normal state on a von Neumann algebra M . For every $C < \infty$, every fixed $z \in M$, and every $\eta > 0$, there is $\delta > 0$ such that*

$$\|D\| \leq C, \quad \|D\|_{\varphi} < \delta \quad \implies \quad \|Dz\|_{\varphi} < \eta. \quad (7.2)$$

The choice of δ is uniform over all such D . For $U, D \in M$ and $b \in M_{\varphi}$,

$$\|UD\|_{\varphi} \leq \|U\| \|D\|_{\varphi}, \quad \|Db\|_{\varphi} \leq \|b\| \|D\|_{\varphi}. \quad (7.3)$$

More generally, the right multiplier bound is uniform when z ranges over a norm-bounded set that is precompact in state norm. Consequently, a bounded family of errors tending uniformly to zero in state norm remains so after right multiplication by a fixed finite word whose factors range over such precompact sets or over bounded subsets of the centralizer. In particular, any specified bounded sequence converging to a fixed operator in state norm can be used as a right factor uniformly over its entire sequence. These conclusions permit bounded strong- $$ approximation of fixed letters in fixed-length scalar words, uniformly in the intervening bounded centralizer factors and in the independently varying indices of other specified approximation sequences.*

Proof. Let Ω be the GNS state vector. Faithfulness makes it separating and hence cyclic for M' . For any $\alpha > 0$, choose $c' \in M'$ with $\|z\Omega - c'\Omega\| < \alpha$. Then every $\|D\| \leq C$ satisfies

$$\|Dz\|_{\varphi} \leq C\alpha + \|Dc'\Omega\| = C\alpha + \|c'D\Omega\| \leq C\alpha + \|c'\| \|D\|_{\varphi}.$$

Choosing α first proves the uniform assertion (7.2), including families indexed by additional parameters.

The left multiplier bound is immediate. For $b \in M_{\varphi}$ and $D_0 \geq 0$, centrality gives $\varphi(b^*D_0b) = \varphi(D_0bb^*)$. The functional $c \mapsto \varphi(D_0c)$ is positive on M_{φ} , since for $c \geq 0$ there it equals $\varphi(c^{1/2}D_0c^{1/2})$. It follows that $\varphi(b^*D_0b) \leq \|b\|^2\varphi(D_0)$, proving Equation (7.3).

For a norm-bounded state-norm precompact set $\mathcal{C}$, take a finite state-norm approximation $z_1, \dots, z_m$ at a prescribed accuracy. If $z \in \mathcal{C}$ is close to z_i , then

$$\|Dz\|_{\varphi} \leq \|Dz_i\|_{\varphi} + C\|z - z_i\|_{\varphi}.$$

The finitely many fixed-factor moduli give one modulus for $\mathcal{C}$. The set consisting of a convergent sequence and its limit is precompact, so this applies to every bounded sequence z_j with $\|z_j - z\|_\varphi \rightarrow 0$. Apply the right multiplier bounds successively in the order in which the factors occur. At each step the errors remain uniformly bounded in operator norm, so the next modulus continues to apply. For a scalar word with a middle error D , the estimate

$$|\varphi(UDV)| \leq \|U\| \|DV\|_\varphi$$

and telescoping reduce the approximation to the preceding bounds. Strong-* approximation ensures state-norm convergence of a replaced letter and its adjoint whenever both occur. For each fixed approximant the finite right words may be expanded as needed; its approximation index can then be sent to infinity after the uniform error estimates. $\square$

Proposition 7.4 (Transfer of fixed scalar moments). *Let $A \subset M$, φ , and $\mathcal{K}$ be as in Lemma 7.1. Fix a finite symbol size s , the corresponding cylinder algebra $\mathcal{Y}$, and the generating partition sequence $(\mathcal{P}_j)$ just described. For any fixed integer $k \geq 1$ and fixed factors $Z_1, \dots, Z_k$, each chosen in M or in $\mathcal{Y}$, define $Z_h^{(j)} = Z_h$ for a factor chosen in M and $Z_h^{(j)} = \text{ev}_{\mathcal{P}_j}(Z_h)$ for a factor chosen in $\mathcal{Y}$. Use the same independent symbols for all evaluations in a word. Then*

$$\lim_{j \rightarrow \infty} \mathbb{E}_{\{z_a\}} \varphi(Z_1^{(j)} \cdots Z_k^{(j)}) = \Phi(Z_1 \cdots Z_k). \quad (7.4)$$

Proof. We first record how operator approximation will be used. We then reduce to centered cyclic words. Their label averages are sums of terms with prescribed equality relations among partition indices. A shortest pair of linked occurrences will give a state Cauchy–Schwarz estimate whose small factor is controlled by Proposition 3.2.

Uniform approximation in fixed words. Lemma 7.3 controls bounded errors in the state norm when the right word is formed from fixed operators and bounded varying A -functions. Expanding each fixed finite normalizer sum when needed, telescoping and

$$|\varphi(UDV)| \leq \|U\| \|DV\|_\varphi$$

bound scalar word errors uniformly over the evaluated diagonal functions. If a fixed N -letter is approximated by a bounded strongly convergent sequence from N_{alg} , that sequence and its limit form a set precompact in the state norm, so the lemma applies to the whole sequence as right factors. Each normalizer sum is expanded only after that approximant is fixed; no uniform bound on all such expansions is needed. The order is to fix an approximant, pass to the partition limit, and then let its approximation error tend to zero. The same statements apply in the model, with Φ and bounded factors from B , which lies in its centralizer.

Reduction to centered alternating words. We induct on the number of factors. A word of length one in M gives an identity. For one factor in $\mathcal{Y}$, average its cylinder coefficients. At almost every base point, distinct sites in a fixed finite translated list eventually belong to distinct atoms of $\mathcal{P}_j$. The averaged coefficient thus converges boundedly almost everywhere to its conditional label average. This is also state-norm convergence, and the fixed normalizers are allowed right multipliers. It proves Equation (7.4) in length one.

Adjacent factors from the same algebra can be folded into one factor. For an alternating word, subtract and add the N -expectation of each factor in the model. Make this same expansion on the evaluated side; it is not necessary that a centered model factor remain centered after evaluation.

The expectations of $\mathcal{Y}$ -factors lie in N_{alg} and are fixed by evaluation. Every term containing an expectation can be reduced in length by folding it into a neighbor. If an arbitrary N -element from the expectation of an M -factor must be multiplied into a $\mathcal{Y}$ -factor, first approximate it boundedly in the strong-* topology by N_{alg} . Such approximation follows from generation and Kaplansky density, and is legitimate on both sides by the uniform word estimates above. The induction hypothesis therefore handles all terms except fully centered alternating words. Their model expectation is zero by freeness.

We may approximate each centered M -factor boundedly in the strong-* topology by an entire analytic element for σ^φ belonging to M° . For example, convolve its modular orbit with Gaussian approximate identities; complex translations of the Gaussian give the entire continuation. The expectation E_N commutes with the modular group, so the convolutions remain centered. Their complex modular translates also remain in M° , by analytic continuation after applying E_N .

Each centered $\mathcal{Y}$ -factor is a finite sum of terms vb , where v is a base kernel groupoid normalizer and b is a cylinder function whose conditional label average is zero. To obtain this form, move each coefficient to the right of its normalizer by covariance on the relevant supports, incorporating bounded base coefficients into b , and subtract the expectation of each summand. The sum of the subtracted expectations is zero. Cut the initial support of each v as in Section 3 to make it entire analytic. These cuts preserve centering of the cylinder coefficients and have arbitrarily small state errors in the fixed words. The cylinder functions and their evaluations belong to the respective centralizers. Hence complex modular shifts of vb are compatible with evaluation: the shift only changes the cut normalizer by the same bounded base formula in both algebras. We fix these analytic approximants before passing to the partition limit; their complex-shift norms may enter all subsequent bounds.

For an odd alternating word of length at least three, the first and last factors lie in the same constituent algebra. Apply the KMS identity

$$\varphi(ZY) = \varphi(Y\sigma_{-i}^\varphi(Z))$$

to the first evaluated analytic factor, and the corresponding identity for Φ in the model. Fold the last factor with the shifted first factor and apply the induction hypothesis to the shorter word. Its model value equals that of the original centered word and is therefore zero.

For an even alternating word, rotate in the same way if necessary to start in M° . Expand the other-side factors as vb and absorb each v into the preceding centered M -factor. Modular shifts of normalizers, if present, only add fixed base coefficients. Writing the length of this word as $2m$, it remains to prove that the averaged evaluations of

$$\varphi(y_1 d_1 \cdots y_m d_m) \tag{7.5}$$

tend to zero. Here the y_h are fixed bounded entire analytic elements of M° , and each $d_h \in A$ is the substitution in a centered finite cylinder function. All constants below may depend on this fixed word and on the fixed analytic approximants.

Fourier expansion and index equalities. Split each cylinder domain according to the domains of its translated sites and the equality pattern among those sites. On each piece, its distinct label symbols are independent. Fourier expansion on the finite product of cyclic groups expresses the centered coefficient as a finite sum of monomials with fixed bounded A -coefficients. Centering removes the constant Fourier term. Delete occurrences having exponent zero modulo s . Each remaining monomial has at least one occurrence, and all its exponents are nonzero modulo s . Within

each diagonal block d_h , the sites of its occurrences are pointwise distinct on that block's indicated support.

We may, after further finite splitting, regard every translation used at an occurrence as an automorphism of the full abelian algebra A . Here the extension away from the specified support need not belong to $\mathcal{R}$. Proposition 3.2 shows that the y_h annihilate the atomic part of A on both sides, so only diffuse supports contribute to Equation (7.5). The partial isomorphisms map these supports into the diffuse part modulo null sets. Split a nonnull common domain piece of a block into two positive-measure pieces. On either smaller piece, both its domain and the range of each restricted map have nonnull diffuse complements in the diffuse part: the other piece and its image provide such complements. Normalize the measures on each pair of complements separately and use the isomorphism theorem for atomless standard probability spaces [Kec95, Theorem 17.41]. It extends each restricted map to a measure-class automorphism of the diffuse part. Use the identity on the atomic part. This gives finitely many terms, and for an occurrence o we can write its translation as

$$\lambda_o(f) = f \circ \gamma_o, \quad f \in A,$$

where γ_o is a nonsingular measurable bijection modulo null sets.

Expand Equation (7.5) in partition indices a_o , one for each occurrence. Averaging the independent symbols z_a keeps precisely the index equality patterns for which the sum of the Fourier exponents in each equality class is zero modulo s . Since individual exponents are nonzero, every such class has at least two occurrences. Inclusion–exclusion over the finitely many possible coarsenings removes inequalities between distinct classes. Thus it suffices to prove vanishing for a sum with prescribed equalities within the classes of a partition of the occurrences, every class having size at least two, and with no inequalities imposed between classes. An occurrence contributes the projection $\lambda_o(e_{a_o})$.

We will keep selected pair equalities as explicit summation indices. Whenever the selected pairs form a forest in each equality class, extend them to a spanning tree of that class. Encode the other tree edges with independent auxiliary phases. For an edge between indices a, b , its equality indicator is

$$\mathbf{1}_{\{a=b\}} = \mathbb{E}_\zeta(\zeta_a \overline{\zeta_b}),$$

where the ζ_a are independent uniform circle phases; use a fresh phase vector for each edge. At an occurrence o , let $w_o(a)$ be the product of its incident auxiliary edge phases, with conjugates according to their orientations and with empty product one. Thus $|w_o(a)| = 1$. Set

$$U_o = \sum_a w_o(a) \lambda_o(e_a).$$

The projections $\{\lambda_o(e_a)\}_a$ form a partition, so U_o is a diagonal unitary. Summing an unretained index gives U_o , while at a retained index j one has

$$w_o(j) \lambda_o(e_j) = U_o \lambda_o(e_j).$$

Thus each U_o , as an operator, is independent of the retained numerical indices. Different U_o may be correlated through the auxiliary phases; the estimates below hold uniformly over their joint realizations as diagonal unitaries before averaging. Products of these factors commute within each diagonal block.

A pair in one diagonal block. Suppose two occurrences in one equality class belong to the same block. Retain their pair equality and encode the remaining tree edges as just described. Summing the pair index produces the projection

$$p\mathcal{P} = \mathbf{1}_{\text{support}} \sum_a \lambda_o(e_a) \lambda_{o'}(e_a).$$

The two sites are distinct on the fixed support, while the partitions separate points. Hence $\mu(p_{\mathcal{P}}) \rightarrow 0$. Use KMS rotation to place this projection first in the word. Only fixed analytic letters are shifted; diagonal factors are in the centralizer and remain unchanged. The rotated word has the form $\varphi(p_{\mathcal{P}}Z)$ with a uniform norm bound on Z . Centrality gives

$$|\varphi(p_{\mathcal{P}}Z)| = |\varphi(p_{\mathcal{P}}Zp_{\mathcal{P}})| \leq \|Z\| \mu(p_{\mathcal{P}}),$$

so these sums tend to zero uniformly in the auxiliary phases.

A shortest pair between different blocks. It remains to treat equality patterns with at most one occurrence of any class in each diagonal block. This case has $m \geq 2$. Number the blocks by $\mathbb{Z}/m\mathbb{Z}$, and for distinct positions h, k define

$$d_m(h, k) = \min\{(k - h) \bmod m, (h - k) \bmod m\},$$

where both residues are taken in $\{1, \dots, m-1\}$. Among pairs of occurrences in the same class, choose one minimizing this cyclic distance, and call the minimum ℓ . Choose the endpoint order so that one shortest arc runs forward in the cyclic order of the word, and label its blocks $0, 1, \dots, \ell$; either endpoint order is allowed in an antipodal tie.

For each interior block $h = 1, \dots, \ell-1$, choose one occurrence and one other occurrence of its class. Its mate lies outside the entire chosen arc, including the endpoints. Indeed, a mate in another block k on that arc would satisfy $d_m(h, k) \leq |h - k| < \ell$, contradicting minimality. The same inequality shows that the interior classes are distinct from each other and from the endpoint class. When $\ell = 1$ there are no interior blocks.

Retain the endpoint pair and these interior-to-mate pairs. These pairs lie in distinct equality classes, so they form a forest and can be included in the spanning trees used above. The retained indices are $J = (j_0, j_1, \dots, j_{\ell-1})$, with j_0 linking the endpoints. Write

$$E_0 = \lambda_0(e_{j_0}), \quad E'_\ell = \lambda_\ell(e_{j_0}), \quad E_h = \lambda_h(e_{j_h}) \quad (1 \leq h < \ell).$$

Each j_h for $h \geq 1$ occurs in one further retained projection in an exterior block. Several such mate projections may lie in the same exterior block.

Rotate the summands by KMS to begin at E_0 , and insert E_0 at the far end inside the state, using centrality. Absorb fixed diagonal coefficients into neighboring fixed letters. More explicitly, within a diagonal block place the retained projections first, then the auxiliary unitaries, then its fixed coefficient; absorb that coefficient into the following intervening letter. The fixed intervening letters, including any modular translates created by rotation, remain bounded analytic elements of M° . Relabel them as y_h , and write $V_h \in A$ for the auxiliary unitaries, which are independent of the explicit indices. Split the word at E'_ℓ using $(E'_\ell)^2 = E'_\ell$. Before averaging the auxiliary unitaries, the resulting summands have the form $\varphi(X_J Y_J)$, where

$$\begin{aligned} X_J &= E_0 V_0 \left(\prod_{h=1}^{\ell-1} y_h E_h V_h \right) y_\ell E'_\ell, \\ Y_J &= E'_\ell V_\ell B_{j_1, \dots, j_{\ell-1}} E_0. \end{aligned} \tag{7.6}$$

The product in X_J is in increasing order and is omitted when $\ell = 1$. In $B_{j_1, \dots, j_{\ell-1}}$, each exterior mate projection for these indices occurs exactly once; every other factor is independent of them. Diagonal factors in the same block commute.

Figure 1 illustrates the selected links in one allowed cyclic pattern.

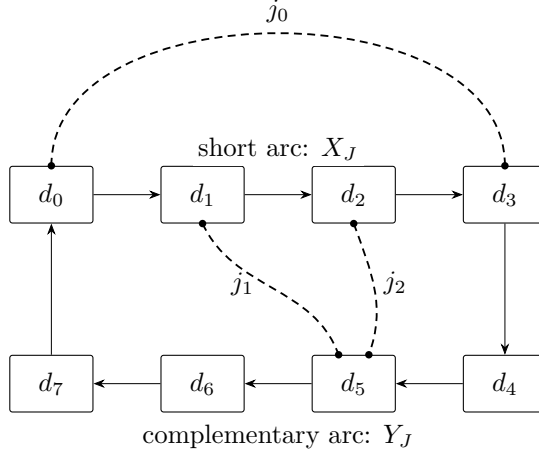


Figure 1: Selected equality links in one fixed cyclic word with eight diagonal blocks and $\ell = 3$. The short arc $0 \rightarrow 1 \rightarrow 2 \rightarrow 3$ and its complement indicate the positions contributing to X_J and Y_J . The two marks in block 5 are distinct mate occurrences. Further occurrences and equality links are omitted; distinct prescribed classes may use equal numerical partition indices after inequalities between classes have been removed.

State Cauchy–Schwarz gives

$$\left| \sum_J \varphi(X_J Y_J) \right| \leq \left(\sum_J \varphi(X_J X_J^*) \right)^{1/2} \left(\sum_J \varphi(Y_J^* Y_J) \right)^{1/2}. \quad (7.7)$$

The second square sum is uniformly bounded. First drop E_ℓ' from the positive operator bound for $Y_J^* Y_J$. Order the exterior mate occurrences from left to right, using commutativity for those in one diagonal block. Write $\mathbf{j} = (j_1, \dots, j_{\ell-1})$. At the first remaining mate occurrence, factor

$$B_{\mathbf{j}} = C_0 F_j B_{\mathbf{j}'}',$$

where j is that occurrence's index, $\mathbf{j}'$ lists the others, $\{F_j\}_j$ is its projection partition, and $C_0, B_{\mathbf{j}'}'$ are independent of j . Then

$$\sum_j B_{\mathbf{j}}^* B_{\mathbf{j}} = B_{\mathbf{j}'}'^* \left(\sum_j F_j C_0^* C_0 F_j \right) B_{\mathbf{j}'}' \leq \|C_0\|^2 B_{\mathbf{j}'}'^* B_{\mathbf{j}'}'.$$

Iterating over the remaining mate occurrences, with the fixed norm bounds on the intervening multipliers, gives

$$\sum_{j_1, \dots, j_{\ell-1}} B_{j_1, \dots, j_{\ell-1}}^* B_{j_1, \dots, j_{\ell-1}} \leq C 1$$

for a fixed constant C . When $\ell = 1$, this is simply the fixed norm bound for the exterior word. Finally sum the E_0 corners in the state. As $\{\lambda_0(e_{j_0})\}_{j_0}$ is a partition in the centralizer, this leaves a bound independent of $\mathcal{P}$ and of all auxiliary unitaries.

We have reduced the moment estimate to the first square sum in Equation (7.7). Its smallness will come from a collision between the two endpoint indices. The interior pinchings first converge at fixed coarse endpoint data; only afterward will those coarse endpoint data be refined.

Remove V_0 from the state by centrality and sum the interior indices in $X_J X_J^*$. The result is

$$\sum_{j_0} \varphi(\lambda_0(e_{j_0}) C_{\mathcal{P}}(\lambda_\ell(e_{j_0})) \lambda_0(e_{j_0})), \quad (7.8)$$

where the positive map $C_{\mathcal{P}} : A \rightarrow M$ is given by the nested operations

$$\begin{aligned} C_{\mathcal{P}}(f) &= L_1, \\ L_\ell &= y_\ell f y_\ell^*, \\ L_h &= y_h S_{\lambda_h(\mathcal{P})}(V_h L_{h+1} V_h^*) y_h^* \quad (h = \ell - 1, \dots, 1). \end{aligned}$$

For $\ell = 1$ this means simply $C_{\mathcal{P}}(f) = y_1 f y_1^*$.

Fix a coarse partition $\mathcal{Q}$ from the generating sequence and let $\mathcal{P}$ refine it. Positivity permits each input projection in Equation (7.8) to be enlarged to its coarse piece. Regrouping the output corners by centrality then bounds that expression by

$$\sum_{e \in \mathcal{Q}} \varphi(\lambda_0(e) C_{\mathcal{P}}(\lambda_\ell(e)) \lambda_0(e)). \quad (7.9)$$

For each fixed $f \in A$, the nested expression defining $C_{\mathcal{P}}(f)$ converges in state norm to the expression in which each pinching is replaced by E_A and the V_h disappear. This convergence is uniform over the auxiliary unitaries. Indeed, work from the innermost pinching outward. Each translated sequence $\lambda_h(\mathcal{P}_j)$ is generating, so Lemma 2.3 gives its state-norm convergence to E_A on a fixed input. Conjugation by V_h commutes with the pinching, preserves the state norm, and fixes the limiting A -valued datum. Pinching is contractive in the state norm. Its error can thus be propagated through the fixed left and right multipliers y_h, y_h^* using Lemma 7.3, uniformly over all the remaining phases. This proves the asserted inside-out convergence.

Define positive kernel maps on A by

$$K_h(f) = E_A(y_h f y_h^*) \quad (1 \leq h \leq \ell).$$

For each fixed input $f = \lambda_\ell(e)$, $e \in \mathcal{Q}$, the preceding convergence applies. Since $\mathcal{Q}$ is finite, the limit of Equation (7.9), with $\mathcal{Q}$ fixed, is

$$\sum_{e \in \mathcal{Q}} \int_X \lambda_0(e) (K_1 \circ \dots \circ K_\ell)(\lambda_\ell(e)) d\mu. \quad (7.10)$$

Proposition 3.2, applied to the adjoints of the y_h , says that these kernels and their composition are atomless almost everywhere. In taking compositions, a preceding kernel avoids the fixed null exceptional set where a following kernel may fail to be atomless.

Finally refine $\mathcal{Q}$ along the generating sequence. The integrand sum in Equation (7.10) tests whether γ_0 of the output site and γ_ℓ of the input site lie in the same partition piece. These conditions decrease to equality of those two points. Since γ_ℓ is one-to-one, this is a graph having zero conditional mass for each atomless row of the composed kernel. Nonsingularity makes all fixed null exceptions irrelevant. The kernel masses are bounded, so bounded convergence shows that Equation (7.10) tends to zero.

For each fixed coarse $\mathcal{Q}$, the limsup of the first square sum, even after taking the supremum over the auxiliary unitaries at each fine $\mathcal{P}$, was bounded by Equation (7.10). First take that fine-partition limsup and then refine $\mathcal{Q}$. The first square sum therefore vanishes uniformly in the auxiliary phases. The second square sum was uniformly bounded, so Equation (7.7) proves vanishing of the equality-pattern sum. Averaging the auxiliary phases, undoing the finite inclusion-exclusion and Fourier expansions, and then removing the fixed analytic approximations proves the required vanishing in Equation (7.5). The induction now proves Equation (7.4). $\square$

The moment-transfer statement uses a fixed finite symbol alphabet. The following approximation allows it to be applied to the measurable color partitions supplied by Theorem 6.1.

Lemma 7.5 (Cylinder approximation of projection partitions). *In the Bernoulli extension of Lemma 7.1, let $p_1, \dots, p_r \in B$ be projections with $\sum_i p_i = 1$ and $p_i p_h = 0$ for $i \neq h$. For every $\delta > 0$ there is a projection partition $b_1, \dots, b_r \in B$, each of whose members is a finite cylinder function for one common finite dyadic symbol size, such that*

$$\|b_i - p_i\|_\Phi < \delta \quad (1 \leq i \leq r).$$

Every evaluation of this cylinder partition by the preceding finite-symbol substitution gives a projection partition of 1 in A , modulo the fixed null sets of its definition.

Proof. The extension σ -algebra is generated by base measurable sets and countably many label-coordinate readings along the partial graphings. Dyadic approximations to the label coordinates therefore show that the finite cylinder set algebra, allowing arbitrary base measurable sets, is dense in probability. Approximate each of the measurable sets corresponding to the p_i by sets in this algebra. Disjointize the approximations in order and put the remaining complement into one piece, retaining exactly r pieces and covering the identity. The resulting symmetric-difference measures can be made arbitrarily small, which gives the displayed state-norm bounds for their projections. There are only finitely many pieces and coordinates, so all of them use one common finite dyadic symbol size after refinement.

The resulting identities $b_i = b_i^* = b_i^2$, $b_i b_h = 0$ for $i \neq h$, and $\sum_i b_i = 1$ are identities of finite cylinder functions. The full-support argument in the construction of ev_P shows that substitution preserves all these identities. Thus their evaluations form a genuine projection partition in A . $\square$

8 Completion of the paving argument

We now combine the coloring theorem and moment transfer. The coloring estimates apply to finite-support approximations of a Cartan operator. Before transferring a fixed moment, we must return to the original operator without requiring a rate of approximation depending exponentially on the moment order. A clipping argument accomplishes this. Its use in a nonracial state requires the following elementary observation about varying right multipliers.

8.1 Continuous calculus for varying sequences

Lemma 8.1. *Let $\mathcal{L}$ be a von Neumann algebra with a faithful normal state ω . Let $\mathcal{R}_\omega$ consist of all uniformly bounded sequences (R_n) in $\mathcal{L}$ such that*

$$\|D_n R_n\|_\omega \longrightarrow 0$$

whenever (D_n) is uniformly bounded and $\|D_n\|_\omega \rightarrow 0$. Then $\mathcal{R}_\omega$ is an algebra, closed for the supremum operator norm. It contains all constant sequences and all uniformly bounded sequences in the centralizer of ω . It is also closed under adding uniformly bounded sequences converging to zero in $\|\cdot\|_\omega$.

If $(U_n), (V_n) \in \mathcal{R}_\omega$ are self-adjoint, uniformly bounded, and $\|U_n - V_n\|_\omega \rightarrow 0$, then for every continuous function f on a compact interval containing their spectra,

$$(f(U_n)), (f(V_n)) \in \mathcal{R}_\omega, \quad \|f(U_n) - f(V_n)\|_\omega \longrightarrow 0.$$

Proof. The assertions for constant sequences and centralizer sequences follow from Lemma 7.3. If $(R_n), (S_n) \in \mathcal{R}_\omega$ and (D_n) is a bounded state-norm-null sequence, then $(D_n R_n)$ is another such sequence, so $\|D_n R_n S_n\|_\omega \rightarrow 0$. Addition is immediate. Closure in the supremum norm follows by approximating the right multiplier uniformly before taking the limit in n .

For the perturbation assertion, if (E_n) is bounded and state-norm-null, then

$$\|D_n E_n\|_\omega \leq \|D_n\| \|E_n\|_\omega \rightarrow 0.$$

Thus $(R_n + E_n) \in \mathcal{R}_\omega$ whenever $(R_n) \in \mathcal{R}_\omega$.

For a polynomial f , the assertion about membership follows from the algebra property. Polynomial differences are controlled by the telescoping identity

$$U_n^j - V_n^j = \sum_{h=0}^{j-1} U_n^h (U_n - V_n) V_n^{j-1-h}.$$

Every right multiplier on the right belongs to $\mathcal{R}_\omega$, while the left multipliers are uniformly bounded. Each summand therefore tends to zero in state norm. Uniform polynomial approximation on the common spectral interval proves both assertions for continuous f . $\square$

8.2 Paving in the presence of a central state

Proposition 8.2. *Let $A \subseteq M$ be a masa and let φ be a faithful normal state on M with A in its centralizer. For $0 < \varepsilon < 1$, put*

$$r_e = \left\lceil 20000^2 \varepsilon^{-2} \right\rceil.$$

For every self-adjoint contraction $x \in M$ and every $\rho > 0$, there are a partition $P = (p_1, \dots, p_{r_e})$ in A and a projection $q \in M$ such that

$$\varphi(1 - q) < \rho, \quad \|q(S_P(x) - E_A x)q\| \leq \varepsilon.$$

Here E_A is the φ -preserving conditional expectation onto A .

Proof. By Lemma 2.4, it suffices to work with separable predual. The conditional expectation in that reduction is the restriction of E_A , so proving the stated diagonal choice in the smaller inclusion proves it in the original one. We henceforth use the notation of Sections 3 and 7.

Fix $r = r_e$ and write

$$a = E_A x, \quad y = E_N x - E_A x \in N, \quad w = x - E_N x \in \ker E_N.$$

The three elements are self-adjoint, $\|a\| \leq 1$, and $\|y\|, \|w\| \leq 2$. Moreover $E_A y = 0$: uniqueness of the φ -preserving expectation onto A gives $E_A E_N = E_A$.

Finite-support Cartan approximations. Choose self-adjoint contractions in N_{alg} converging strongly to $E_N x$. To obtain them, first apply Kaplansky density to the norm closure of the finite normalizer algebra, then approximate its self-adjoint contractions in norm by self-adjoint elements of N_{alg} and rescale. Bounded strong approximation can be taken sequentially in the faithful separable state representation.

For each approximant, only finitely many partial isomorphisms occur in its support. We may also arrange a bound on their edge Radon–Nikodym ratios. Indeed, for such a finite list of maps γ , remove from X each domain set where $\Delta(\gamma v, v)$ lies outside $[b^{-1}, b]$, and let $e_b \in A$ be the

indicator of the remaining set. As $b \rightarrow \infty$, these projections increase strongly to 1. Compressing the approximant by e_b preserves self-adjointness and its contraction bound, and restricts every surviving edge to the required ratio range. Enlarge the finite list by its inverses if necessary. Choosing b sufficiently large for each approximant preserves strong convergence.

Subtracting the diagonal now gives a sequence

$$y_n = y_n^* \in N_{\text{alg}}, \quad E_A y_n = 0, \quad \|y_n\| \leq 2, \quad \|y_n - y\|_\varphi \rightarrow 0. \quad (8.1)$$

Each $y_n/2$ is represented by measurable zero-diagonal Hermitian contractions on orbit ℓ^2 spaces, supported on an undirected graph of finite degree with bounded edge cocycle ratios. These bounds may depend on n . Theorem 6.1 permits this dependence: it does not affect r .

One partition for the two operator terms. Use the fixed relative Bernoulli extension $\tilde{N}$ and amalgamated product

$$\mathcal{K} = M *_N \tilde{N}, \quad \Phi = \varphi|_N \circ E_N^{\mathcal{K}},$$

from Lemma 7.1. Apply Theorem 6.1 to $y_n/2$ with

$$t = \frac{4}{\sqrt{r}}, \quad \alpha = \frac{\varepsilon}{32}, \quad m = n, \quad \eta = \left(\frac{\varepsilon}{16}\right)^{2n}. \quad (8.2)$$

Denote the color projections by $p_1^n, \dots, p_r^n \in B$. They form a partition, belong to the centralizer of Φ , and satisfy

$$\tilde{E}_N(p_i^n) = r^{-1}1 \quad (1 \leq i \leq r). \quad (8.3)$$

With $\beta = 80$, the norm bound supplied by the coloring theorem, after restoring the factor two, obeys

$$2(2\beta t + \alpha) = \frac{1280}{\sqrt{r}} + \frac{\varepsilon}{16} \leq \frac{253}{2000}\varepsilon < \frac{\varepsilon}{4}. \quad (8.4)$$

Put

$$\hat{Y}_n = \sum_{i=1}^r p_i^n y_n p_i^n \in \tilde{N}.$$

At a root where the radius- n ball has the asserted pinching bound, $\hat{Y}_n^n$ applied to the root vector agrees with the n th power of that principal matrix applied to the same vector. Every path of length n starting at the root stays in its radius- n ball. At all other roots, $\|\hat{Y}_n\| \leq 2$. Integrating the squared norms of these vectors gives

$$\Phi(\hat{Y}_n^{2n}) \leq \left(\frac{\varepsilon}{4}\right)^{2n} + 2^{2n} \left(\frac{\varepsilon}{16}\right)^{2n}. \quad (8.5)$$

This uses diagonal integration in the relation representation, not a trace on the nonsingular relation algebra.

For the original operators set

$$Y_n = \sum_{i=1}^r p_i^n y p_i^n, \quad W_n = \sum_{i=1}^r p_i^n w p_i^n.$$

Pinching by a centralizer partition is contractive in the state norm, so Equation (8.1) implies

$$\|Y_n - \hat{Y}_n\|_\Phi \rightarrow 0. \quad (8.6)$$

By Lemma 7.2 and Equation (8.3),

$$\|W_n\| \leq 2 \left(\frac{1}{r} + \frac{2}{\sqrt{r}} \right) \leq \frac{40001}{200000000}\varepsilon < \frac{\varepsilon}{4}. \quad (8.7)$$

Here orthogonality of the p_i^n makes the norm of the block sum the maximum of its block norms. Thus the same r projections control the complementary term; there is no second partition refinement.

Clipping before fixing a transferred moment. Let $f : \mathbb{R} \rightarrow [-\varepsilon/2, \varepsilon/2]$ be the continuous clipping function

$$f(s) = \max\{-\varepsilon/2, \min\{s, \varepsilon/2\}\}.$$

Spectral calculus and Equation (8.5) show that

$$\Phi(1_{\{|\hat{Y}_n| > \varepsilon/2\}}) \leq \left(\frac{2}{\varepsilon}\right)^{2n} \Phi(\hat{Y}_n^{2n}) \leq 2^{-2n} + 4^{-2n}.$$

Since $\|\hat{Y}_n\| \leq 2$, it follows that

$$\|\hat{Y}_n - f(\hat{Y}_n)\|_{\Phi} \longrightarrow 0. \quad (8.8)$$

We apply Lemma 8.1 in $\mathcal{K}$. Each sequence (p_i^n) lies in $\mathcal{R}_{\Phi}$, and r is fixed. Hence (Y_n) and (W_n) belong to that algebra, being finite sums of products of centralizer sequences and fixed operators. Equation (8.6) also places $(\hat{Y}_n)$ in it. Equations (8.6)–(8.8) and the continuous-calculus conclusion of the lemma yield

$$\|Y_n - f(Y_n)\|_{\Phi} \longrightarrow 0. \quad (8.9)$$

The self-adjoint operators

$$Z_n = Y_n + W_n = \sum_{i=1}^r p_i^n (x - a) p_i^n$$

therefore differ by a bounded state-norm-null sequence from

$$Q_n = f(Y_n) + W_n, \quad \|Q_n\| \leq \frac{3\varepsilon}{4}.$$

Both sequences belong to $\mathcal{R}_{\Phi}$. For every fixed integer $k \geq 1$, telescoping their powers gives

$$\limsup_{n \rightarrow \infty} \Phi(Z_n^{2k}) \leq \left(\frac{3\varepsilon}{4}\right)^{2k}. \quad (8.10)$$

Only a fixed power is used at this stage. In particular, Equation (8.1) required no prescribed rate of convergence.

Transfer and the final spectral cut. Fix $\rho > 0$. First choose k so large that $(3/4)^{2k} < \rho/4$. Then choose n sufficiently large in Equation (8.10) that

$$\Phi(Z_n^{2k}) < \frac{\rho}{2} \varepsilon^{2k}.$$

With n and k fixed, approximate the partition (p_i^n) in B by a finite-symbol cylinder partition, using Lemma 7.5. Bounded state-norm approximation of these self-adjoint projections is strong-* approximation, so the fixed polynomial moment changes arbitrarily little. We may retain the strict upper bound $\rho \varepsilon^{2k}$ with positive slack.

Proposition 7.4 applies to the expansion of this moment. For a sufficiently fine base partition, its averaged evaluated moment remains strictly less than $\rho \varepsilon^{2k}$. Every evaluation is a genuine projection partition in A , and its even moment is nonnegative. Some deterministic evaluation consequently gives a partition $P = (p_1, \dots, p_r)$ with

$$\varphi \left(\left[\sum_{i=1}^r p_i (x - a) p_i \right]^{2k} \right) < \rho \varepsilon^{2k}.$$

Let $z = \sum_i p_i(x - a)p_i = z^*$ and take $q = 1_{[-\varepsilon, \varepsilon]}(z) \in M$. Then

$$\varphi(1 - q) \leq \varepsilon^{-2k} \varphi(z^{2k}) < \rho, \quad \|qzq\| \leq \varepsilon.$$

Because $a \in A$ commutes with the partition, $z = S_P(x) - a$. This proves the proposition with the number of colors fixed before ρ . $\square$

8.3 Arbitrary inclusions

Proof of Theorem 1.1. If $x = 0$, take $p_1 = 1$, $p_2 = \cdots = p_{r_\varepsilon} = 0$, $a = 0$, and $q = 1$. Otherwise replace x by $x/\|x\|$. Proposition 8.2, together with Proposition 2.2, supplies the required paving in the original representation using

$$r = \left\lceil 400000000\varepsilon^{-2} \right\rceil + \lceil 4/\varepsilon \rceil + 1$$

projections. This number depends only on ε . The supporting corner and all partitions, diagonal elements, and compression projections may depend on F and δ , as permitted in Definition 1.1. The construction has diagonal norm at most one, and

$$r \leq 400000000\varepsilon^{-2} + 4\varepsilon^{-1} + 3 \leq 400000007\varepsilon^{-2} < 500000000\varepsilon^{-2}.$$

Multiplying the diagonal by the original $\|x\|$ restores scale and gives every assertion of Theorem 1.1. $\square$

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